

# Drought in the United States: Science, Policy, and Selected Federal Authorities

Updated October 2, 2026

Congressional Research Service

<https://crsreports.congress.gov>

R46911



# Drought in the United States: Science, Policy, and Selected Federal Authorities

**R46911**

October 2, 2026

**Charles V. Stern,**  
**Coordinator**  
Specialist in Natural  
Resources Policy

**Eva Lipiec, Coordinator**  
Specialist in Natural  
Resource Policy

For a copy of the full report,  
please call 7-5700 or visit  
[www.crs.gov](http://www.crs.gov).

Drought occurs to some extent almost every year in areas of the United States. Droughts consist of a deficiency of moisture that results in adverse effects, with some drought types developing over weeks or months and others developing more quickly (i.e., a *flash drought*). Droughts can simultaneously reduce available water supplies and increase water demands. Drought has the potential to affect economic and environmental conditions on local, regional, and national scales, as well as to disrupt water supplies for households, communities, and other users. Although droughts are generally a common component of climate variability that may be seasonal, multiyear, or multi-decadal in duration, variable precipitation and rising temperatures are intensifying droughts in some regions.

Since the federal government generally defers to state primacy in surface and groundwater allocation, states and local entities typically lead efforts to prepare for drought and manage water resources during droughts. Multiple federal agencies contribute to efforts to predict, plan for, and respond to drought. The federal government, and in particular the National Oceanic and Atmospheric Administration, plays a key role in researching, monitoring, and sharing information about drought through the National Integrated Drought Information System and the U.S. Drought Monitor. Other federal agencies, such as the U.S. Department of Agriculture (USDA) and the U.S. Geological Survey, also research and monitor drought factors and conditions.

USDA also provides the primary federal financial aid to lessen the impacts of drought on agriculture and compensate for agricultural production loss after drought onset. Federal programs administered by USDA and other federal agencies may assist with drought in their support of other water infrastructure investments. For example, Congress has authorized programs supporting nonfederal and tribal efforts to develop water conservation, water reuse and recycling, rural water supplies, or other municipal and industrial water supplies. Some of these programs may prioritize projects that lessen the impacts of drought even if these programs do not focus exclusively on drought.

Congress has authorized federal assistance for other aspects of drought, but these programs generally are limited in scope. Federal water resource agencies such as the U.S. Bureau of Reclamation (Reclamation) and the U.S. Army Corps of Engineers (USACE) may face difficult tradeoffs in operating federal water projects during drought. Both agencies have the authority to conduct activities to mitigate drought impacts. Congress has directed both Reclamation and USACE to plan for future droughts and to incorporate forecast-informed operations into operations at federally authorized projects.

Recent severe droughts, including widespread drought conditions in some areas of the western United States, have raised the public profile of drought and led to congressional and administrative proposals to prepare for and respond to its impacts. Recent congresses have approved supplemental funding for drought activities (e.g., in the Infrastructure Investment and Jobs Act [P.L. 117-58]). Congressional interest in drought during the 119<sup>th</sup> Congress has included legislation to provide new and amended authorities for drought planning and response; emergency appropriations to alleviate drought impacts and enhance related activities; and oversight of ongoing federal drought science, preparedness, and management efforts.

# Contents

|                                                                           |    |
|---------------------------------------------------------------------------|----|
| Introduction .....                                                        | 1  |
| Overview of Drought in the United States .....                            | 1  |
| What Is Drought? .....                                                    | 2  |
| Drought Trends.....                                                       | 4  |
| Observed Trends .....                                                     | 5  |
| Future Projections .....                                                  | 6  |
| Drought Classification .....                                              | 6  |
| Drought Prediction for the United States .....                            | 10 |
| Federal Drought Policy and Coordination.....                              | 12 |
| Coordination.....                                                         | 12 |
| Selected Federal Authorities.....                                         | 13 |
| Monitoring and Research .....                                             | 14 |
| USDA Drought Support Programs for Farmers and Ranchers.....               | 16 |
| Direct Payments .....                                                     | 16 |
| Insurance .....                                                           | 18 |
| Cost-Share Assistance .....                                               | 19 |
| Loans.....                                                                | 19 |
| Conservation .....                                                        | 20 |
| NOAA Drought Support Programs .....                                       | 21 |
| Drought and Federal Water Resource Facilities and Projects .....          | 22 |
| New and Expanded Water Storage Projects.....                              | 24 |
| Drought Operations Manuals and Planning.....                              | 26 |
| Forecast-Informed Reservoir Operations.....                               | 29 |
| Drought Flexibilities and the Endangered Species Act.....                 | 31 |
| Support for Nonfederal Drought Planning and Projects .....                | 32 |
| Bureau of Reclamation Authorities.....                                    | 32 |
| USDA Programs and Authorities .....                                       | 34 |
| USACE Authorities.....                                                    | 36 |
| U.S. Environmental Protection Agency Programs .....                       | 37 |
| Selected Federal Drought Programs for Tribes.....                         | 40 |
| Recent Supplemental Funding Responding to Drought .....                   | 43 |
| Infrastructure Investment and Jobs Act.....                               | 44 |
| Extending Government Funding and Delivering Emergency Assistance Act..... | 44 |
| Inflation Reduction Act of 2022.....                                      | 45 |
| Disaster Relief Supplemental Appropriations Act, 2023.....                | 46 |
| Disaster Relief Supplemental Appropriations Act, 2025.....                | 46 |
| Proposed Drought Authorities in the 119 <sup>th</sup> Congress .....      | 46 |
| Coordination.....                                                         | 46 |
| Monitoring and Research .....                                             | 46 |
| NOAA Drought Support Programs .....                                       | 47 |
| Reclamation Programs .....                                                | 47 |
| USACE Programs .....                                                      | 48 |
| EPA Programs .....                                                        | 48 |
| USDA Programs.....                                                        | 49 |

## **Figures**

|                                                                                                                          |    |
|--------------------------------------------------------------------------------------------------------------------------|----|
| Figure 1. Generalized Cascade of Drought Response and Recovery Across Water Sources .....                                | 3  |
| Figure 2. Example of a U.S. Drought Monitor Map.....                                                                     | 8  |
| Figure 3. Percentage of United States Land Area in U.S. Drought Monitor Categories .....                                 | 9  |
| Figure 4. Percentage of the Northeast and West Region Land Areas in U.S. Drought Monitor Categories .....                | 9  |
| Figure 5. Example Illustrating Widespread and Regional Drought in the Continental United States .....                    | 10 |
| Figure 6. Examples of U.S. Monthly and Seasonal Drought Outlooks in the United States.....                               | 11 |
| Figure 7. Bureau of Reclamation and U.S. Army Corps of Engineers Dams with Water Supply and/or Irrigation Purposes ..... | 23 |
| Figure 8. Depiction of Atmospheric River Originating Near Hawaii and Extending to the U.S. West Coast.....               | 30 |

## **Tables**

|                                                                                       |    |
|---------------------------------------------------------------------------------------|----|
| Table 1. Selected USACE Water Supply and Drought Authorities Enacted Since 2020 ..... | 25 |
|---------------------------------------------------------------------------------------|----|

## **Contacts**

|                         |    |
|-------------------------|----|
| Author Information..... | 49 |
|-------------------------|----|

## Introduction

Drought occurs to some extent almost every year in areas of the United States (see “What Is Drought?”). Droughts consist of a deficiency of moisture that results in adverse effects, with some drought types developing over weeks or months and others developing quickly (i.e., a *flash drought*). Drought has the potential to create economic and environmental impacts on local, regional, and national scales, as well as disruptions in water supplies for households, communities, and other users. Although droughts are generally a common component of climate variability that may be seasonal, multiyear, or multi-decadal in duration, variable precipitation and rising temperatures are intensifying droughts in some regions (see “Drought Trends”). Severe and widespread droughts in parts of the United States since 2012 have raised the profile of drought and led to increasing congressional and administrative proposals to prepare for and respond to its impacts.

Multiple federal agencies contribute to efforts to predict, plan for, and respond to drought. The National Oceanic and Atmospheric Administration (NOAA) plays a key role in monitoring, predicting, and sharing information about drought, as well as leading the federal interagency National Integrated Drought Information System (NIDIS) and codeveloping the U.S. Drought Monitor with the University of Nebraska–Lincoln and the U.S. Department of Agriculture (USDA). USDA and the U.S. Geological Survey (USGS) also research and monitor drought factors and conditions. USDA provides the primary federal financial aid to lessen drought’s impacts and compensate for agricultural production loss after its onset. Federal water resource agencies such as the U.S. Bureau of Reclamation (Reclamation) and the U.S. Army Corps of Engineers (USACE) may face difficult tradeoffs in operating federal water projects during drought; both agencies have the authority to conduct activities to mitigate drought impacts. Various other federal agencies also have a role in drought response and mitigation.

This report provides an overview of drought in the United States, including information on drought science, monitoring, and forecasts, and on drought types and intensity classifications. It discusses federal authorities related to drought planning and response, with a focus on selected water-related agricultural, environmental, and natural resource authorities with explicit ties to drought. It does not discuss broader disaster-related authorities and their potential nexus to drought, such as the programs and authorities of the Federal Emergency Management Agency (FEMA), and interactions between drought and other hazards and concerns (e.g., wildfire, dust, and public health).

## Overview of Drought in the United States<sup>1</sup>

The following sections provide information about drought in the United States, including a discussion of the different types of drought, trends in recent decades, potential linkages between drought and climate change now and in the future, and how the impacts of drought are classified and forecasted in the United States.

---

<sup>1</sup> For more information, contact Eva Lipiec, Specialist in Natural Resources Policy, or Nicole T. Carter, Specialist in Natural Resources Policy.

## What Is Drought?

Drought can be difficult to define.<sup>2</sup> Scientists have defined several types of drought, including meteorological, hydrological, agricultural, socioeconomic, and ecological.<sup>3</sup> Meteorological drought, the degree of dryness, in comparison to a “normal” or average amount of dryness, and the duration of a dry period, can sometimes lead to the other types of drought.<sup>4</sup> For instance, periods of low precipitation (in comparison to the average amount) can result in shortfalls of surface or subsurface water supply, known as hydrological drought.

Some experts use the term *drought cascade* to describe how changes in precipitation and snowpack may propagate through various water sources (**Figure 1**).<sup>5</sup> Scientists have also characterized some meteorological droughts as *hot drought* or *flash droughts*. In contrast to precipitation-driven droughts, hot droughts are a result of high air temperatures, as warmer air absorbs more water than cooler air.<sup>6</sup> Some scientists use the term *flash drought* to broadly identify droughts that are characterized by unusually rapid intensification, either at the onset or the exacerbation of an existing drought.<sup>7</sup> Flash drought is usually set in motion by precipitation deficits, accompanied by anomalously high temperatures, winds, and/or solar radiation.<sup>8</sup>

Higher demand for water for human activities and vegetation in areas of limited water supply increases the severity of certain types of drought. For example, drought during the growing season is often considered more severe—in terms of impacts—than similar conditions when cropland lies dormant. For policy purposes, drought often becomes an issue when it results in a water supply deficiency. Drought can lead to water restrictions affecting municipal and industrial supplies, decreased hydropower generation and power plant cooling efficiency, navigation limitations and disruptions, harm to drought-sensitive ecosystems and species, and increased fire and heat wave risk, among other effects.<sup>9</sup>

Scientific advances have enhanced understanding of drought and the forces that contribute to it, although debates remain on those forces’ relative importance. Drought at a specific location often results from a mix of various interacting atmospheric conditions, land-atmosphere feedbacks, topographic features, and human activity, such as water use, among other factors. When certain spatially and temporally large, geographic-scale air and ocean conditions persist, droughts may

<sup>2</sup> National Integrated Drought Information System (NIDIS), “Defining Basics,” <https://www.drought.gov/what-is-drought/drought-basics>. Hereinafter NIDIS, “Defining Drought.”

<sup>3</sup> NIDIS, “Defining Drought.”

<sup>4</sup> National Drought Mitigation Center (NDMC), “Types of Drought,” <https://drought.unl.edu/Education/DroughtIn-depth/TypesofDrought.aspx>.

<sup>5</sup> For example, Dr. Zachary H. Hoyleman, “The Drought Cascade,” July 24, 2026, presentation slides, Montana Climate Office, [https://www.eesi.org/files/Zach\\_Hoyleman\\_072426\\_Slides.pdf](https://www.eesi.org/files/Zach_Hoyleman_072426_Slides.pdf).

<sup>6</sup> Some scientists use the term *temperature-dominated drought* rather than hot drought (NOAA, Climate.gov, “Beyond 2016: Year in Review,” January 4, 2017, <https://www.climate.gov/news-features/blogs/beyond-data/beyond-2016-year-review>; and U.S. Global Change Research Program (USGCRP), “Water,” in *Fifth National Climate Assessment*, 2023). For examples of research on hot droughts see Bradley Udall and Jonathan Overpeck, “The Twenty-First Century Colorado River Hot Drought and Implications for the Future,” *Water Resources Research*, vol. 53, no. 3 (February 17, 2017); Peter T. Soule and Paul A. Knapp, “The evolution of ‘Hot’ droughts in Southern California, USA from the 20<sup>th</sup> to the 21<sup>st</sup> century,” *Journal of Arid Environments*, vol. 220 (February 2024); and Yizhou Zhuang et al., “Anthropogenic warming has ushered in an era of temperature-dominated droughts in the western United States,” *Science Advances*, vol. 10, no. 45 (November 6, 2024).

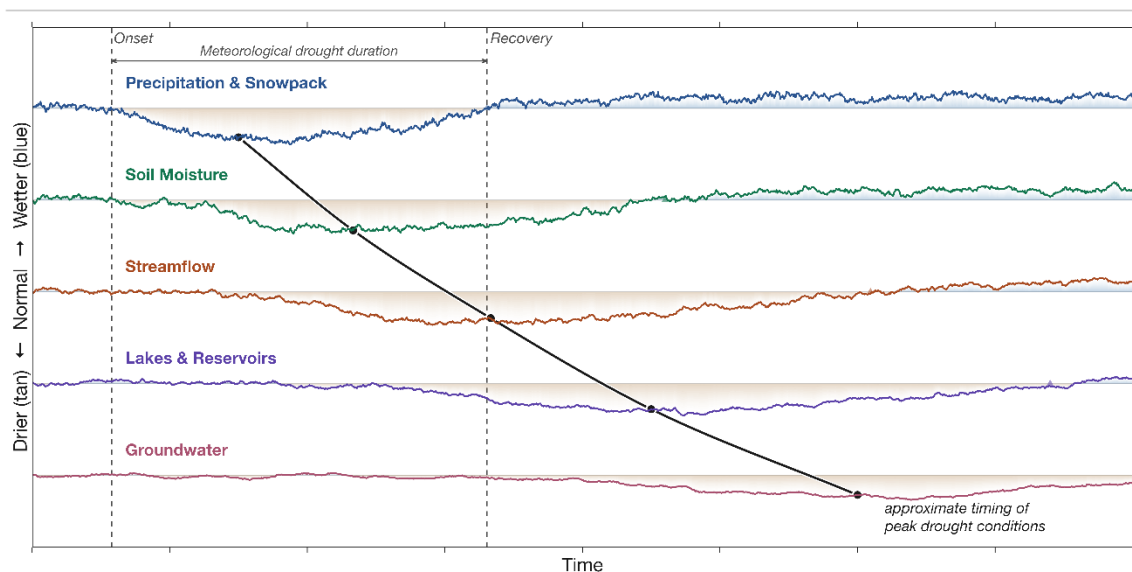
<sup>7</sup> Jordan I. Christian et al., “Flash Drought: A State of the Science Review,” *WIREs Water*, vol. 11, no. 3 (January 9, 2024).

<sup>8</sup> NIDIS, “What Is Drought: Flash Drought,” <https://www.drought.gov/what-is-drought/flash-drought>.

<sup>9</sup> NIDIS, “Defining Drought”; and USGCRP, “Water,” in *Fifth National Climate Assessment*, 2023.

become prolonged. For example, the periodic ocean conditions known as the El Niño-Southern Oscillation (ENSO) and the Pacific Decadal Oscillation (PDO) can impact atmospheric patterns for several weeks to several months to several consecutive years.<sup>10</sup> (For more on the role Pacific Ocean conditions play in droughts in the United States, see text box “Droughts and the El Niño-Southern Oscillation.”)

**Figure 1. Generalized Cascade of Drought Response and Recovery Across Water Sources**



**Source:** Dr. Zachary H. Hoylman, Montana Climate Office, University of Montana, July 27, 2026. Modified by CRS.

**Note:** The term *drought cascade* described how changes in precipitation and snowpack may propagate through various water sources and features. Areas shaded in tan show drier-than-normal conditions, and those shaded in blue show wetter-than-normal conditions. The black points and line denote the “peak” drought conditions for each of the features, illustrating how drought conditions can persist in water sources even after precipitation and snowpack return to normal conditions.

<sup>10</sup> The El Niño-Southern Oscillation (ENSO) is a periodic fluctuation in sea surface temperature (El Niño) and the air pressure of the overlying atmosphere (Southern Oscillation) across the equatorial Pacific Ocean (NOAA, National Centers for Environmental Information (NCEI), “El Niño/Southern Oscillation (ENSO),” <https://www.ncei.noaa.gov/access/monitoring/enso/>). For more information about the Pacific Decadal Oscillation, see NOAA, NCEI, “Pacific Decadal Oscillation (PDO),” <https://www.ncei.noaa.gov/access/monitoring/pdo/>.

### Droughts and the El Niño-Southern Oscillation

The El Niño-Southern Oscillation (ENSO) is a set of central and eastern tropical Pacific Ocean conditions—particularly sea surface temperatures (SSTs), air pressure, and winds—that can affect the path of the jet stream over mid-latitude North America and other regions globally, thereby influencing regional temperatures and precipitation. ENSO conditions occur periodically and can last months to years. They often are described by phase: El Niño, neutral, or La Niña. The El Niño phase refers to warmer-than-average SSTs in the central tropical Pacific region, and the La Niña phase refers to cooler-than-average SSTs in the same region. Scientists have increasingly linked drought in portions of the United States to SST anomalies in the Pacific Ocean. Some findings include the following:

- Broadly, El Niño may cause areas in the northern United States and Canada to be drier and warmer than usual, whereas La Niña may cause drought in the southern United States.
- The strength, timing, and duration of La Niña and El Niño conditions are among the various factors (e.g., topography) affecting how and where drought conditions do and do not occur in the United States.
- More specifically, El Niño is associated with higher likelihood of below-average precipitation and above-average temperatures in the northern United States during December through May. La Niña is associated with higher likelihood of below-average precipitation and above-average temperatures in the southern contiguous United States in September through May. The most consistent dry pattern is found in the Southwest (California, Nevada, Arizona, and Utah). Other areas also see an increased chance of below-average precipitation, such as the Southeast, Texas, New Mexico, and Oklahoma in December through May and the central Great Plains in March through May.
- The effects of El Niño and La Niña on drought conditions on Hawaii and the insular areas of the Pacific vary within the region.

Regarding teleconnections patterns that may contribute to drought, a 2021 Intergovernmental Panel on Climate Change (IPCC) report on climate change stated that it is “virtually certain” that ENSO will remain the dominant mode of variability on interannual time scales (i.e., preferred spatial pattern and temporal year-to-year variation). IPCC also found it “very likely” that ENSO-related rainfall variability would intensify in the future.

**Sources:** National Integrated Drought Information System, “The El Niño-Southern Oscillation and Drought Outlook in the United States,” September 8, 2025, <https://www.drought.gov/news/el-nino-southern-oscillation-and-drought-outlook-united-states-2025-09-08>; U.S. Global Change Research Program, “Hawai’i and U.S.-Affiliated Pacific Islands” in *Fifth National Climate Assessment*, 2023; and IPCC, “Chapter 4: Future Global climate: Scenario-based Projections and Near-term Information,” *Climate Change 2021: The Physical Science Basis. Working Group I Contribution to the IPCC Sixth Assessment Report*, 2021.

**Notes:** IPCC authors use terms of likelihood as a quantitative measure of uncertainty in a finding, expressed probabilistically. For instance, virtually certain outcomes have a 99%-100% assessed likelihood of occurring, while very likely outcomes have a 90%-100% likelihood (IPCC, “Chapter 1: Framing, Context and Methods, Box 1.1,” *Climate Change 2021: The Physical Science Basis. Working Group I Contribution to the IPCC Sixth Assessment Report*, 2021).

## Drought Trends

Both the U.S. Global Change Research Program (USGCRP) and the Intergovernmental Panel on Climate Change (IPCC) periodically evaluate available data on drought trends and potential future projections in the United States and North America. USGCRP, an interagency program established by Congress to coordinate and integrate federal research on global change,<sup>11</sup> released a periodic national climate assessment in 2023, known as the Fifth National Climate Assessment

<sup>11</sup> For more information about USGCRP, see CRS Report R48478, *U.S. Global Change Research Program (USGCRP): Overview and Considerations for Congress*, by Kathryn G. Kynett.

(NCA5).<sup>12</sup> In 2021, the Intergovernmental Panel on Climate Change (IPCC) released its sixth assessment on the underlying physical science of climate change.<sup>13</sup>

## Observed Trends

USGCRP, in its 2023 assessment, found that

- Drought is complex and dependent on fluctuations in moisture supply, evaporation, and ecosystem and land surface processes.<sup>14</sup> Observed changes were “highly season- and region-dependent over the U.S., especially in the West.”
- Western North America experienced several decadal-scale droughts in the 20<sup>th</sup> century and multi-decadal “megadroughts” prior to 1600 AD, and, as of 2023, an ongoing multi-decadal megadrought.<sup>15</sup>
- The U.S. Southwest was “the only region in which the total area of unusually dry soil moisture is increasing.”<sup>16</sup> Southwest droughts were lasting longer and reflective of a long-term aridification trend, rather than temporary extreme events.
- Much of the United States is “vulnerable to rapid-onset flash droughts,” and evidence exists that these events are drying out soil more quickly as the world warms.<sup>17</sup>

The IPCC in its 2021 assessment indicated

- inconsistent meteorological drought trends, leading to low confidence, for both western and eastern North America and medium confidence in a decrease in the duration and frequency of meteorological droughts in central North America.<sup>18</sup>
- mixed signals in or inconsistent trends of agricultural and ecological drought, and therefore low confidence, in central and eastern North America and medium

---

<sup>12</sup> It is unclear whether USGCRP plans to release a subsequent assessment. In August 2026, the Trump Administration made a proposed amendment to the Fifth National Climate Assessment available for public comment (Science and Technology Policy Office, “United States Global Change Research Program; Availability for Public Comment on the Proposed Amendment to the Fifth National Climate Assessment (NCA5),” 91 *Federal Register* 54893, August 25, 2026). The proposed amendment states, “NCA5 is amended to clarify that results dependent on RCP8.5, SSP5-8.5, or equivalent high emissions pathways should not be used as expected, baseline, business-as-usual, likely, central, or policy-relevant federal planning futures. These pathways are now treated in this Amendment as implausible scenario outputs” (USGCRP, *Proposed Amendment to the Fifth National Climate Assessment (NCA5): Implausible High-Emission Scenario Dependence, Gold Standard Science, and Observational Grounding*, August 2026, <https://www.noaa.gov/sites/default/files/2026-08/NCA5-Proposed-Amendment-August-2026.pdf>).

<sup>13</sup> The Intergovernmental Panel on Climate Change (IPCC) plans to release its seventh assessment by 2029.

<sup>14</sup> USGCRP, “3. Earth Systems Processes,” in *Fifth National Climate Assessment*, 2023.

<sup>15</sup> USGCRP, “3. Earth Systems Processes,” in *Fifth National Climate Assessment*, 2023.

<sup>16</sup> USGCRP, “2. Climate Trends,” in *Fifth National Climate Assessment*, 2023.

<sup>17</sup> USGCRP, “2. Climate Trends,” in *Fifth National Climate Assessment*, 2023.

<sup>18</sup> According to IPCC, author teams communicated the degree of certainty in key findings by expressing the qualitative confidence about the validity of a finding and degree of agreement (levels of confidence are expressed using five qualifiers: very low, low, medium, high, and very high) or the quantified measure of uncertainty expressed probabilistically (e.g., exceptionally unlikely, likely, virtually certain, etc.). For instance, the term *low confidence* was used for various reasons, including if there was limited evidence, which may be due to a lack of data or studies, and if there was a lack of agreement on the type of change (e.g., mixed signals). (IPCC, “Chapter 1: Framing, Context and Methods, Box 1.1,” *Climate Change 2021: The Physical Science Basis. Working Group I Contribution to the IPCC Sixth Assessment Report*, 2021).

confidence in an increase in agricultural and ecological drought in western North America.

- mixed signals or limited evidence between subregions, hence low confidence for hydrologic drought in western, central, and eastern North America.<sup>19</sup>

## **Future Projections**

In regard to potential future drought conditions in the United States, USGCRP projected in its 2023 assessment that climate change was “expected to amplify drought, primarily through warming-induced increases in evaporative demand and surface water losses, and declines in snowfall and water stored in winter snowpack.”<sup>20</sup> Multi-decadal soil moisture droughts in the Southwest were projected to increase in the future, regardless of global warming level, due to projected decreases in springtime precipitation, earlier snowmelt, and warmer temperatures.<sup>21</sup> Other parts of the United States were not projected to aridify to the same extent as the Southwest; the risk of drought in those areas depends on changes to the amount, type, and timing of precipitation and evapotranspiration compared with past trends.<sup>22</sup> Under 1.5°C of global warming, the IPCC report found low confidence in changes across drought types and the three regions (western, central, and eastern North America), with the exception of medium confidence in an increase in agricultural and ecological drought in central North America. The confidence levels in changes under 2°C of global warming were more mixed across the drought types and regions. IPCC found low confidence in changes across all drought types in the eastern region, in meteorological and hydrological drought in the central region, and in meteorological drought in the western region. The report found medium confidence in increased agricultural and ecological drought in the western and central regions and in hydrological drought in the western region.<sup>23</sup>

## **Drought Classification**

Certain measures, often referred to as *drought indicators*, are used to assess and classify the intensity and type of drought. Local, state, and federal entities may use different indicators or sets of indicators, often combined with expert opinions from the academic, public, and private sectors, to classify drought.

The U.S. Drought Monitor (USDM) is a map created weekly by meteorologists and climatologists from NOAA, USDA, and the nonfederal National Drought Mitigation Center. They use their expertise to interpret multiple indicators, indexes, and stakeholder information to

---

<sup>19</sup> IPCC, “Chapter 11: Weather and Climate Extreme Events in a Changing Climate,” in *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, 2021, Table 11.21, pp. 11-229 to 11-231 (hereinafter, report cited as IPCC, *AR6 Physical Science Basis*). For a depiction of the areas included in the western, central, and eastern North America designations, see IPCC, “Chapter 1: Framing, Context, and Methods,” in *AR6 Physical Science Basis*, p. 1-197.

<sup>20</sup> USGCRP, “3. Earth Systems Processes,” in *Fifth National Climate Assessment*, 2023. Evaporative demand is defined as moisture that the atmosphere pulls from the land surface through evaporation and plant transpiration.

<sup>21</sup> USGCRP, “2. Climate Trends,” in *Fifth National Climate Assessment*, 2023.

<sup>22</sup> USGCRP, “2. Climate Trends,” in *Fifth National Climate Assessment*, 2023.

<sup>23</sup> IPCC, “Chapter 11: Weather and Climate Extreme Events in a Changing Climate, Table 11.21” in *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, 2021. The IPCC report also includes confidence levels for changes under 4°C of global warming, associated with “a ‘worst-case’ scenario with no mitigation.” Those findings are not discussed here.

estimate the intensity and effects of ongoing drought conditions (**Figure 2**).<sup>24</sup> Several dozen unique indicators are used to create the USDM map, but not all geographic areas are represented equally by all of the indicators.<sup>25</sup> The USDM intensity scheme—D0 (abnormally dry), D1 (moderate), D2 (severe), D3 (extreme), and D4 (exceptional)—depicts broad-scale conditions but not necessarily drought circumstances at the local scale. For example, the regions depicted as red in **Figure 2** faced extreme drought conditions for the week preceding June 30, 2026, but they may have contained local areas and individual communities that experienced less or more severe drought. The estimated drought intensity can be a trigger for local, state, and federal responses to drought.

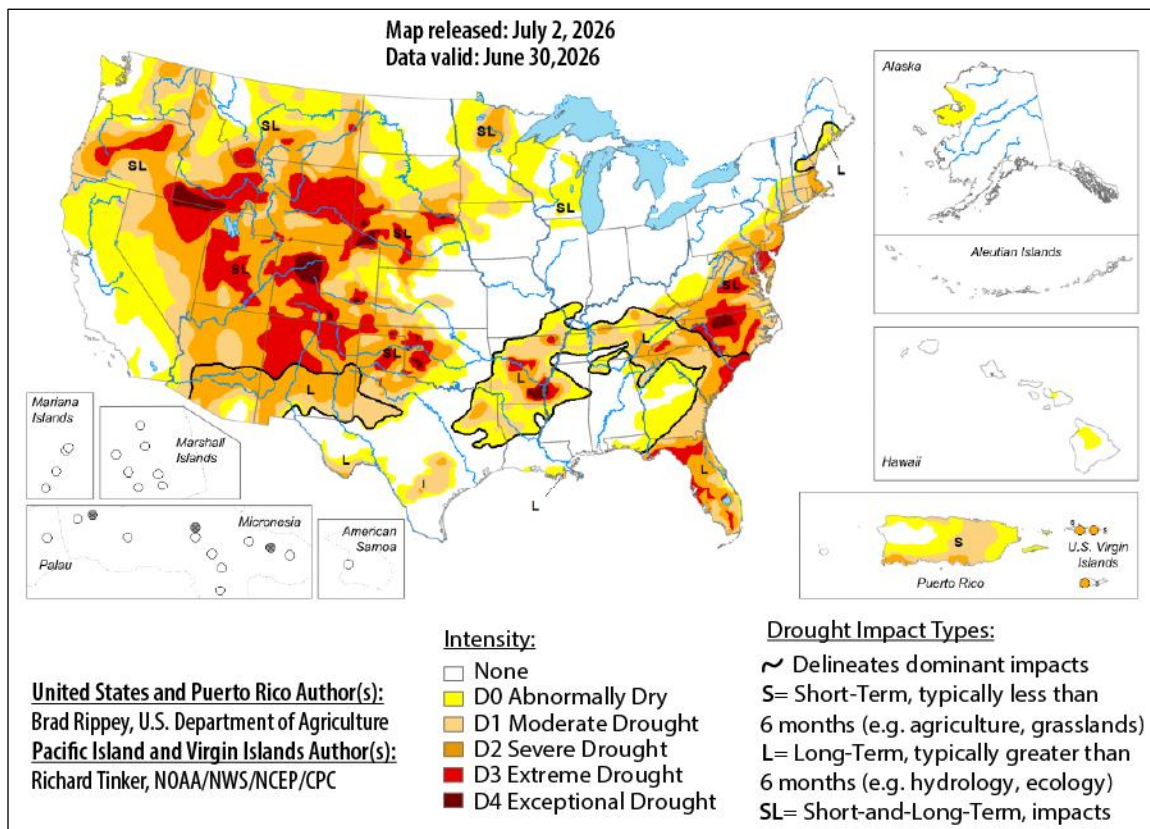
In addition to the color-coded D0-D4 designations, USDM maps often include “S” and “L” designations to provide additional information about the nature of drought across broad geographic regions (**Figure 2**). The “S” designation indicates the existence of short-term effects on agriculture and grasslands, typically less than six months, and can rapidly change.<sup>26</sup> The “L” designation indicates the existence of long-term effects, such as to hydrology and ecology, typically more than six months, that can persist even with precipitation in the short term. As **Figure 2** shows, some regions of the United States include both an “S” and an “L” designation, indicating that in June 2026, those regions experienced both short- and long-term impacts.

---

<sup>24</sup> For more information, see NMDC, “What Is the USDM?,” <https://droughtmonitor.unl.edu/About/WhatistheUSDM.aspx>.

<sup>25</sup> National Drought Mitigation Center (NDMC), “What Data Are Used to Make the Map?,” <https://drought.unl.edu/Education/Tutorials/usdm/section-four.aspx>. For a discussion of drought indicators, see USDM, “Drought Classification,” <https://droughtmonitor.unl.edu/About/AbouttheData/DroughtClassification.aspx>.

<sup>26</sup> U.S. Drought Monitor (USDM), “Drought Classification,” <https://droughtmonitor.unl.edu/About/AbouttheData/DroughtClassification.aspx>.

**Figure 2. Example of a U.S. Drought Monitor Map**

**Source:** U.S. Drought Monitor, <https://droughtmonitor.unl.edu/>. Modified by CRS.

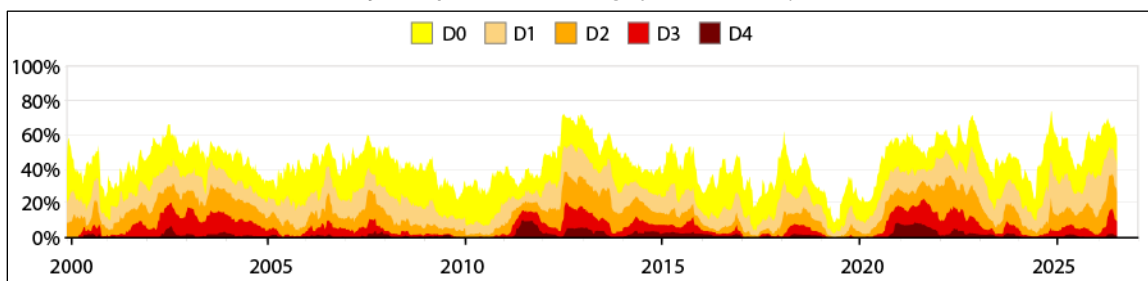
**Note:** U.S. Drought Monitor national maps are updated weekly.

Since USDM began in 2000, some portion of the United States has experienced drought of at least moderate intensity (D1) each year (**Figure 3**).<sup>27</sup> Some portion of the country has experienced *extreme* (D3) or *exceptional* (D4) drought in most years in the same time period.<sup>28</sup> The land area affected by drought can vary widely by year and within a particular year.

<sup>27</sup> USDM, "Time Series," <https://droughtmonitor.unl.edu/DmData/TimeSeries.aspx>.

<sup>28</sup> Extreme (D3) or exceptional (D4) drought did not affect any portion of the country in some time periods since 2000. For example, from January 2000 through early April 2000 and from late March 2019 through mid-May 2019 (USDM, "Time Series," <https://droughtmonitor.unl.edu/DmData/TimeSeries.aspx>).

**Figure 3. Percentage of United States Land Area in U.S. Drought Monitor Categories**  
(January 4, 2000, through June 23, 2026)

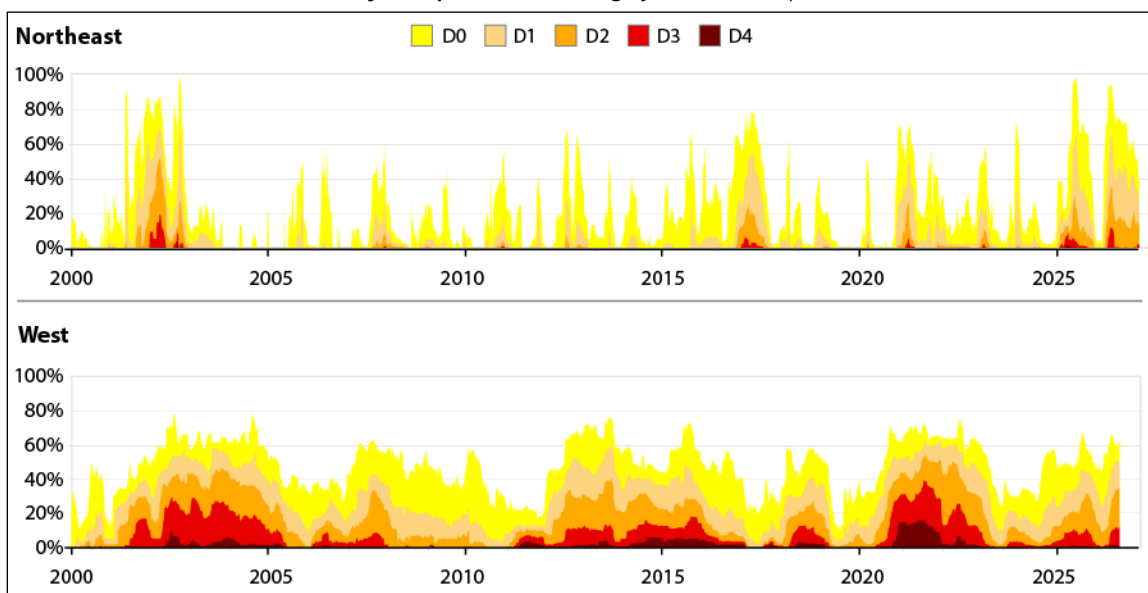


**Source:** U.S. Drought Monitor, "Time Series," <https://droughtmonitor.unl.edu/DmData/TimeSeries.aspx>. Modified by CRS.

**Note:** Includes all 50 states, the District of Columbia, and Puerto Rico. D0 (abnormally dry), D1 (moderate), D2 (severe), D3 (extreme), and D4 (exceptional).

Drought may affect certain regions of the United States on a short-term or a longer-term basis, with varying intensity over time. For example, since USDM began issuing maps in 2000, the Northeast region has rarely experienced extreme (D3) or exceptional drought (D4).<sup>29</sup> In contrast, periods of extreme and exceptional drought have been more common in the West region (Figure 4).<sup>30</sup>

**Figure 4. Percentage of the Northeast and West Region Land Areas in U.S. Drought Monitor Categories**  
(January 4, 2000, through June 23, 2026)



**Source:** U.S. Drought Monitor, "Time Series," <https://droughtmonitor.unl.edu/DmData/TimeSeries.aspx>. Modified by CRS.

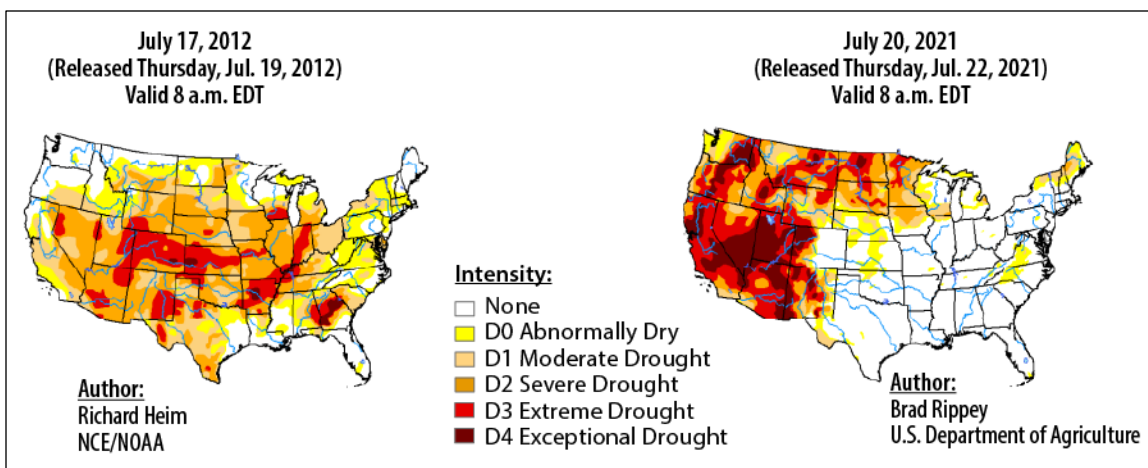
<sup>29</sup> For the purposes of the USDM, the Northeast region includes Connecticut, Delaware, District of Columbia, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont, and West Virginia.

<sup>30</sup> For the purposes of the USDM, the West region includes Arizona, California, Idaho, Montana, Nevada, New Mexico, Oregon, Utah, and Washington.

**Notes:** The Northeast region includes Connecticut, Delaware, District of Columbia, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont, and West Virginia. The West region includes Arizona, California, Idaho, Montana, Nevada, New Mexico, Oregon, Utah, and Washington. D0 (abnormally dry), D1 (moderate), D2 (severe), D3 (extreme), and D4 (exceptional).

Some drought conditions can stretch across the continental United States, whereas others affect smaller areas. For example, on July 17, 2012, abnormally dry or drought conditions (D1-D4) covered roughly 81% of the continental United States, with about 42% of the nation experiencing severe drought or worse (D2-D4; see **Figure 5**). In contrast, on July 20, 2021, about 54% of the continental United States faced abnormally dry or drought conditions (D1-D4), with 38% experiencing severe drought or worse (D2-D4).<sup>31</sup>

**Figure 5. Example Illustrating Widespread and Regional Drought in the Continental United States**



**Source:** U.S. Drought Monitor, “Compare Two Weeks,” <https://droughtmonitor.unl.edu/Maps/CompareTwoWeeks.aspx>. Modified by CRS.

## Drought Prediction for the United States

Reliable drought prediction depends on effectively integrating data on multiple variables, including surface temperature, precipitation, air-sea interactions, topography, soil moisture, land surface processes, and teleconnections, at multiple scales. Scientists seek to understand how these variables interact in order to predict drought onset, severity, and duration.

NOAA periodically releases rapid onset (next two to four weeks), monthly, and seasonal U.S. drought outlooks.<sup>32</sup> According to the agency, the outlooks are based on short-term forecasts and long-range outlooks, regional drought history, and teleconnections.<sup>33</sup> The outlooks depict areas of the United States where NOAA predicts drought is likely to persist, improve, or develop in the next month to several months (**Figure 6**). As of 2026, the USGS also provides current status and forecasts of unusually low river and stream levels, termed streamflow drought, on a scale similar to the USDM (moderate to extreme).<sup>34</sup>

<sup>31</sup> USDM, “Compare Two Weeks,” <https://droughtmonitor.unl.edu/Maps/CompareTwoWeeks.aspx>.

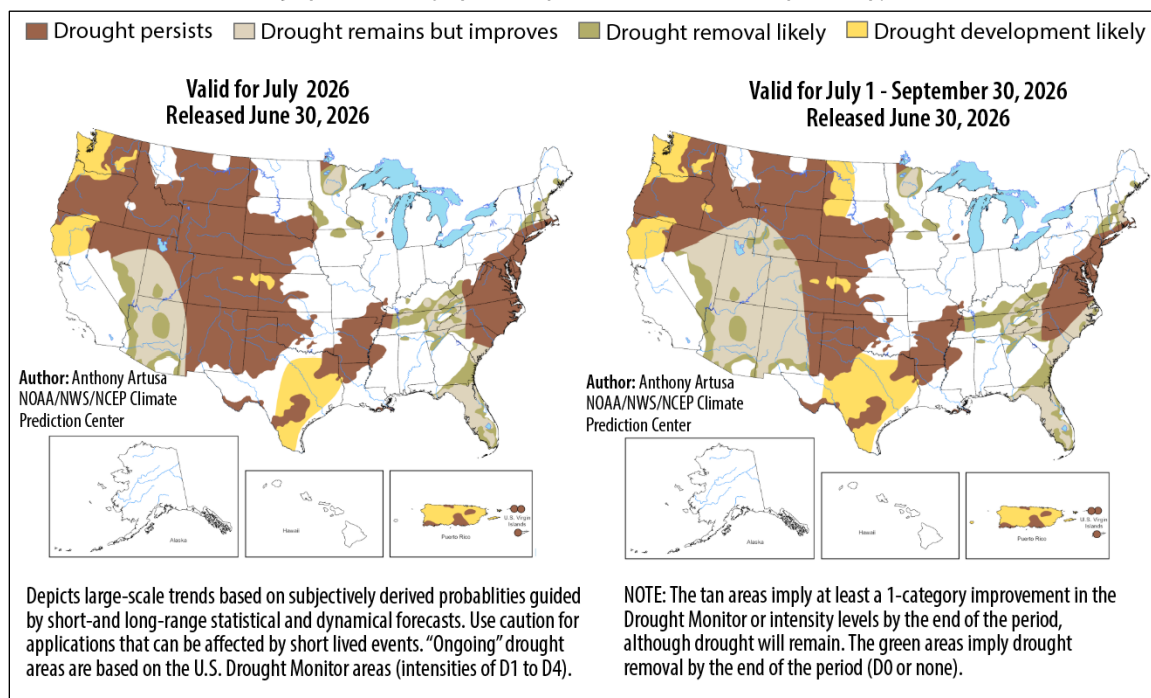
<sup>32</sup> NOAA National Weather Service (NWS), Climate Prediction Center (CPC), “Drought Information,” <https://www.cpc.ncep.noaa.gov/products/Drought/>.

<sup>33</sup> NOAA NWS, “U.S. Seasonal Drought Outlook,” <https://www.weather.gov/safety/drought-ww>.

<sup>34</sup> USGS, “Modeling Streamflow Drought,” <https://water.usgs.gov/vizlab/modeling-drought/>.

**Figure 6. Examples of U.S. Monthly and Seasonal Drought Outlooks in the United States**

(July 2026 and July 1 – September 30, 2026, respectively)



**Source:** NOAA National Weather Service, Climate Prediction Center, "Drought Information," <https://www.weather.gov/safety/drought-ww>. Modified by CRS.

**Note:** Areas shown in white are not experiencing drought conditions.

Drought forecasting remains difficult for various reasons. For example, drought onset forecasting is challenging due, in part, to limitations in accurately predicting precipitation beyond a two-week period.<sup>35</sup> Other factors, such as land cover changes, dam operation, irrigation works, and groundwater extraction, also can confound understanding of hydrologic extremes such as drought.

Another challenge of forecasting future drought conditions is that available drought assessment approaches and indicators rely heavily on historic baselines. The assumption that the statistical characteristics of data relevant to drought (including climate data) are constant through time (i.e., *stationarity*) had enabled scientists to use observed records and statistical methods to anticipate future drought conditions. Some researchers are contemplating other ways to forecast drought, such as using alternative indicators from those currently used, to represent hydrologic conditions and weather and climate phenomena that may deviate from historical patterns (i.e., *nonstationarity*).<sup>36</sup> Other scientists are adopting shorter reference periods for droughts (e.g., most recent 30 years of data), rather than longer historical records.<sup>37</sup>

<sup>35</sup> NIDIS and National Drought Resilience Partnership (NDRP), *Second National Drought Forum, July 30-31, 2019*, Washington, DC, December 2020, p. 13, <https://www.drought.gov/sites/default/files/2020-12/SecondNationalDroughtForumReport.pdf>. Hereinafter, NIDIS and NDRP, *Second National Drought Forum*, 2020.

<sup>36</sup> NASEM, *Improving Future U.S. Drought Assessment*, 2026, Chapters 4 and 6.

<sup>37</sup> Joel Lisonbee et al., "Drought Assessment in a Changing Climate: A Review of Climate Normals for Drought Indices," *Journal of Applied and Service Climatology*, no. 001 (May 30, 2024).

## Federal Drought Policy and Coordination

States generally have primacy in intrastate surface water and groundwater allocation, albeit within the limits of federal supremacy (e.g., federal reserved water rights and/or explicit congressional authorization to control water allocations as a result of compacts between states).<sup>38</sup> As a result, states and local entities typically lead efforts to prepare for and respond to drought. Most states have drought plans in some form, and some of these plans incorporate efforts to reduce drought vulnerabilities.<sup>39</sup> Some states and communities have invested in reducing water demand and expanding drought-resilient supplies (e.g., wastewater reuse and recycling, desalination, and groundwater recharge and management) or have facilitated water banks and markets for water transfers. Community-level drought plans are less common than state plans, except in states that require or support this planning.

In contrast to the state role, federal agencies primarily contribute to efforts to observe, predict, plan for, and respond to drought. Federal financial support to lessen the impacts of drought, after its onset, occurs mostly in the form of financial aid for agricultural production loss. Some federal authorities provide for financial assistance for other aspects of drought (e.g., drought-related planning or projects), but these programs are limited in scope and authority. Other federal agencies have been directed to construct and operate water resources development projects; the operation of these facilities (including their operations plans during drought) may influence the amount of water stored and the delivery of water during drought.

### Coordination

Federal agency actions are coordinated under a variety of mechanisms, including multi-agency partnerships such as the National Integrated Drought Information System (NIDIS), among others. Congress directed NIDIS to focus on drought-related research and communication and tasked it with the following:

- Providing a drought early warning system on a national and regional scale;
- Communicating drought forecasts, conditions, and impacts, such as through USDM, on an ongoing basis to federal and nonfederal entities;
- Providing “timely” data, information, and products on the local, regional, watershed, and state scales;
- Coordinating and integrating federal research and monitoring in support of the early warning system;
- Using existing forecasting and assessment programs and partnerships; and
- Continuing ongoing research and monitoring activities related to drought.<sup>40</sup>

---

<sup>38</sup> Congress has generally required that the federal agencies defer to or comply with state law in the construction and operation of federal facilities pertaining to allocation, control, or distribution of water (see, for example, §8 of the Reclamation Act of 1902, 32 Stat. 390; 43 U.S.C. §§372, 383). However, in some cases Congress has charged the federal government responsibilities that override those general requirements (see, for example, the Boulder Canyon Project Act, (45 Stat. 1057, 43 U.S.C. §617(d)), which directed Reclamation to serve as the sole contractor for Lower Colorado River Basin waters, thereby increasing the federal role in that region relative to other areas).

<sup>39</sup> NDMC collects information on state drought plans; for more information on state plans, see NDMC, “Drought Plans,” <https://drought.unl.edu/Planning/DroughtPlans.aspx>.

<sup>40</sup> 15 U.S.C. 313d.

NIDIS is composed of representatives from federal agencies and nonfederal entities (e.g., nongovernmental organizations and local governments).<sup>41</sup> The group collaborates with additional regional and state partners, as well.<sup>42</sup> Among its activities, NIDIS continues to establish new regional-drought early warning systems to share information and coordinate actions that help communities cope with drought.<sup>43</sup>

Apart from NIDIS, Congress and the executive branch have established other federal coordination mechanisms related to drought. For example, in 2022, Congress established the Interagency Committee on Water Management and Infrastructure (also known as the Water Policy Committee).<sup>44</sup> The committee is to have members from various departments and agencies, and it is to prioritize increasing water resource “drought resiliency” and improving water data management, research, modeling, and forecasting, among other tasks.<sup>45</sup> According to one source, the committee convenes regularly,<sup>46</sup> although CRS was unable to locate information on the contents of those meetings. The executive branch has established other entities, including the federal interagency National Drought Resiliency Partnership (NDRP), via presidential memorandum;<sup>47</sup> CRS did not identify actions taken by the group since 2019.<sup>48</sup>

## Selected Federal Authorities

Congress has enacted a range of authorities related to drought. Some of these authorities deal with drought-related monitoring and research, including early warning and tracking of various drought indicators and conditions. Other authorities involve response to the onset of drought and its effects (e.g., agriculture-related disaster authorities and emergency drinking water supply assistance). Still others focus on longer-term drought response and mitigation. These authorities can take multiple forms, including federal assistance for local and state drought planning, nonfederal water supply projects that increase drought resilience or provide new water supplies, construction of new or expanded federal water storage projects, and the reoperation of existing projects to yield additional water supplies. In some cases, the federal government, at Congress’s direction, also has provided targeted regulatory relief for drought-stricken areas (e.g., relaxation of environmental requirements on pumping water).

Federal drought response authorities regarding monitoring and research, support programs for farmers and ranchers, operations at federal facilities, and nonfederal planning and projects are discussed below. The discussion focuses on agricultural, natural resources, and environmental authorities that directly relate to drought and does not include broader disaster authorities that

---

<sup>41</sup> NIDIS, “Who We Are,” <https://www.drought.gov/about/who-we-are>.

<sup>42</sup> NIDIS, “Partners,” <https://www.drought.gov/about/partners>.

<sup>43</sup> For example, NIDIS launched the Mid-Atlantic drought early warning system in 2026 (NIDIS, “Drought Early Warning System: Mid-Atlantic,” <https://www.drought.gov/dews/mid-atlantic>).

<sup>44</sup> 15 U.S.C. 9708.

<sup>45</sup> 15 U.S.C. 9708.

<sup>46</sup> Interstate Council on Water Policy and Western States Water Council, “Letter to Water Policy Committee, Summer 2026,” July 13, 2026, [https://westernstateswater.org/wp-content/uploads/2026/07/ICWP-WSWC-Letter-to-WPC-Summer-2026\\_Final-1.pdf](https://westernstateswater.org/wp-content/uploads/2026/07/ICWP-WSWC-Letter-to-WPC-Summer-2026_Final-1.pdf).

<sup>47</sup> NIDIS, “Partners,” <https://www.drought.gov/about/partners>. National Drought Resilience Partnership (NDRP) was “institutionalized” by President Obama in a presidential memorandum in 2016 (Executive Office of the President, “Building National Capabilities for Long-Term Drought Resilience,” 81 *Federal Register* 16053, March 25, 2016). The presidential memorandum has not been rescinded.

<sup>48</sup> NDRP, *Priority Actions Supporting Long-Term Drought Resilience*, 2019, <https://www.usda.gov/sites/default/files/documents/ndrp-priority-actions.pdf>.

might be applicable to drought under specific circumstances,<sup>49</sup> or may be available to help mitigate future drought-related damage.<sup>50</sup>

## Monitoring and Research

No single federal agency leads drought monitoring, forecasting, warning, or research, but NOAA, USDA, and USGS play central roles, as coordinated through NIDIS.<sup>51</sup> Multiple federal agencies also work on aspects of drought-related research and monitoring individually and on a multi-agency basis depending on the topic. For example, in 2019, Congress directed the NOAA Administrator to develop a strategy for a national coordinated soil moisture monitoring network.<sup>52</sup> As part of the resultant 2021 strategy, NIDIS, NOAA, USGS, and USDA have supported a joint effort to develop and share near-real-time soil moisture information.<sup>53</sup>

NOAA's drought-related monitoring and research responsibilities fall under several authorities.<sup>54</sup> In 2021, NOAA identified various drought-related activities under these authorities, including observations, modeling, forecasts, products and services, and research for use within the agency, as well as by other federal agencies; tribal, state, and local governments; and individuals. At that time, NOAA's observations and data came from several sources, including historical climate records and reconstructions, geostationary and polar-orbiting weather satellites, and on-the-ground field observations.<sup>55</sup> The resulting products and services contributed to global, national, and regional drought-related indexes, USDM, Drought.gov, and large-scale reports (e.g., National Climate Assessment and IPCC reports), among others.<sup>56</sup> As discussed above, the NOAA-led NIDIS program is the federal government's primary drought-related research and communication

---

<sup>49</sup> For an introduction to these authorities, see CRS Insight IN11696, *Climate Change, Slow-Onset Disasters, and the Federal Emergency Management Agency*, by Diane P. Horn, Erica A. Lee, and Elizabeth M. Webster.

<sup>50</sup> FEMA Hazard Mitigation Assistance programs, such as the Hazard Mitigation Grant Program and the Building Resilient Infrastructure and Communities (BRIC) program, can potentially provide funding for mitigation of drought-related damage. See CRS Insight IN11515, *FEMA Pre-Disaster Mitigation: The Building Resilient Infrastructure and Communities (BRIC) Program*, by Diane P. Horn and CRS Insight IN11733, *Recent Funding Increases for FEMA Hazard Mitigation Assistance*, by Diane P. Horn. Note that FEMA uses the term *mitigation* rather than adaptation, defining mitigation as “any sustained action to reduce or eliminate long-term risk to people and property from natural hazards and their effects.”

<sup>51</sup> NIDIS identified other departments and agencies with roles in drought monitoring, forecasting, warning, and research to include the Department of Commerce's National Institute of Standards and Technology; the Department of Energy; the U.S. Department of Health and Human Services and its Centers for Disease Control and Prevention; the Department of the Interior and its Bureau of Indian Affairs, Bureau of Land Management, National Park Service, U.S. Bureau of Reclamation, and U.S. Fish and Wildlife Service; EPA; NASA; the National Science Foundation; and USACE. Additional agencies participate in other roles. NIDIS, “Partners,” <https://www.drought.gov/about/partners#federal-partners>.

<sup>52</sup> P.L. 115-423.

<sup>53</sup> NIDIS, *A Strategy for the National Coordinated Soil Moisture Monitoring Network*, June 2021, <https://www.drought.gov/documents/strategy-national-coordinated-soil-moisture-monitoring-network>; and NIDIS, “The National Coordinated Soil Moisture Monitoring Network,” <https://www.drought.gov/drought-in-action/national-coordinated-soil-moisture-monitoring-network>.

<sup>54</sup> For example, the National Climate Program Act (P.L. 95-367, as amended), Global Change Research Act of 1990 (P.L. 101-606), Weather Service Modernization Act of 1992 (P.L. 102-567), National Integrated Drought Information System Act of 2006 (P.L. 109-430, as amended), and Weather Research and Forecasting Innovation Act of 2017 (P.L. 115-25, as amended).

<sup>55</sup> NOAA, “NOAA Drought: Science, Observations, and Services,” presentation, July 15, 2021. NOAA was not able to provide a more up to date presentation (Email from NOAA Office of Legislative and Intergovernmental Affairs, July 6, 2026).

<sup>56</sup> NOAA, “NOAA Drought: Science, Observations, and Services,” presentation, July 15, 2021.

hub and serves as a coordinating mechanism for federal drought-related monitoring and research.<sup>57</sup>

USDA also carries out monitoring and research activities related to drought. USDA conducts research and collects data related to drought and water supply on topics such as drought-tolerant crop varieties, evapotranspiration modeling, drought-resilient management techniques and practices, and soil moisture capacity and retention. Many of these research projects and programs are collaborative efforts with other federal and state agencies and with universities. For example, USDA's Snow Survey and Water Supply Forecasting Program (SSWSF) conducts snow surveys and develops water supply forecasts for western states. USDA's NRCS administers the program, using over 900 automated data collection sites in NRCS's Snow Telemetry (SNOTEL) network.<sup>58</sup> The data collected at both SSWSF and SNOTEL sites are used to make water supply forecasts. Another USDA monitoring network, the Soil Climate Analysis Network, takes hourly readings of soil moisture content at over 200 stations.

USGS also supports a wide range of research and monitoring activities. The agency maintains a network of streamgages across the United States, which provides key information for the development of the weekly U.S. Drought Monitor maps and classifications.<sup>59</sup> USGS also has a number of ongoing initiatives to develop new drought-related monitoring and forecasting tools, such as the Next Generation Water Observing System. In 2026, the USGS released River DroughtCast, an interactive online tool that delivers current streamflow drought conditions and weekly forecasts of streamflow drought up to 90 days at approximately 3,000 USGS streamgages.<sup>60</sup> Streamflow drought forecasts can help decision makers anticipate and plan for periods of limited water availability.<sup>61</sup> Another ongoing effort is the USGS Climate Adaptation Science Center's effort to synthesize the "state of the science" on transformational drought (i.e., drought events that can permanently and irreversibly alter ecosystems).<sup>62</sup> As an example of a collaborative effort, NASA, USGS, and a number of stakeholders developed OpenET that employs several methods to generate monthly and annual satellite-based evapotranspiration

---

<sup>57</sup> For more information on NOAA monitoring and research activities, contact Eva Lipiec, Specialist in Natural Resources Policy.

<sup>58</sup> The Snow Survey and Water Supply Forecasting Program is authorized under 26 Stat. 653, as amended, and Reorganization Plan No. IV, as a provision of the Reorganization Act of 1939, as amended (54 Stat. 1234). For more information on this program and the Natural Resources Conservation Service (NRCS's) Snow Survey and Water Supply Forecasting, see <https://www.nrcs.usda.gov/programs-initiatives/sswsf-snow-survey-and-water-supply-forecasting-program>.

<sup>59</sup> See U.S. Drought Monitor, "Drought Classification," <https://droughtmonitor.unl.edu/About/AbouttheData/DroughtClassification.aspx>. During a series of USGS/NIDIS listening sessions in 2022, a general comment relating to streamgages was a desire for a "higher density of gages to capture the "spatial resolution for smaller streams," and also co-locating sensors of different types at the same location as streamgages." NOAA, NIDIS, and USGS, *Drought prediction and water availability: A report on the 2022 USGS-NIDIS National Listening Session Series*, January 23, 2024, <https://www.drought.gov/documents/drought-prediction-and-water-availability-report-2022-usgs-nidis-national-listening>. For retrospective information about USGS streamgages, see CRS Report R45695, *U.S. Geological Survey (USGS) Streamgaging Network: Overview and Issues for Congress*, by Anna E. Normand.

<sup>60</sup> The USGS defines streamflow drought as when streamflow at a particular location drops to levels that are unusually low for the time of year that are identified on the basis of the historical record for each site. USGS, "River DroughtCast," March 12, 2026, <https://www.usgs.gov/tools/river-droughtcast>. To access the tool, see <https://water.usgs.gov/vizlab/streamflow-drought-forecasts>.

<sup>61</sup> USGS, "Modeling streamflow drought," <https://water.usgs.gov/vizlab/modeling-drought/>.

<sup>62</sup> Climate Adaptation Science Centers Project Explorer, "State of the Science Synthesis on Transformational Drought: Understanding Drought's Potential to Transform Ecosystems Across the Country," <https://cascprojects.org/#/project/4f8c64d2e4b0546c0c397b46/5d40ac2fe4b01d82ce8d9db0>.

estimates at the field scale.<sup>63</sup> In 2025, the OpenET team created the Farm and Ranch Management Support (FARMS) tool, the third phase of OpenET's Data Explorer tool, created with features specifically to assist farm operators.<sup>64</sup>

## **USDA Drought Support Programs for Farmers and Ranchers<sup>65</sup>**

Although many factors (e.g., pest infestation, flooding, hail) can pose major production challenges to farmers, drought is the most significant agricultural risk in the United States in terms of production and income loss.<sup>66</sup> Due to the complex way in which plants and livestock respond to heat and water availability, the effect of drought on crops and livestock can be highly variable depending on what is being produced, how, and where. For example, dryland farms with shallow soils may experience a significant loss in production during a moderate drought, whereas irrigated production with ample groundwater may experience negative impacts only during an extreme drought.<sup>67</sup> Soil structure and soil moisture retention, as well as access to ground or surface water, also can heavily influence the level of agricultural production loss from drought.

Congress has authorized support for farmers and ranchers to manage drought risk, pay for losses caused by drought, and incentivize adaptive measures. USDA administers these federal assistance programs, which include direct payments for loss, subsidized insurance, cost-sharing to rehabilitate damaged lands, loans, and financial and technical assistance to implement conservation practices. Most programs have permanent authorization and aim to assist producers recovering from production, financial, and physical loss related to or caused by natural disasters, such as drought. Each program has a different administrative process for producers to request assistance. The loan programs require a disaster declaration or designation for eligibility (discussed further in "Loans," below).<sup>68</sup> The section below includes discussion of some additional types of USDA programs. Supplemental funding for drought-related agricultural loss is discussed in the below section "Recent Supplemental Funding Responding to Drought."

### **Direct Payments**

Some USDA programs provide direct payments to cover production losses above normal mortality. Advance sign-up and fees are not required to participate; however, application deadlines exist following a qualified loss. These programs are permanently authorized and

---

<sup>63</sup> The primary inputs include data from Landsat, Sentinel-2, GOES, and other satellites; weather station networks and models; and field boundary and crop type datasets. The initiative is working toward daily evapotranspiration estimates. For more information, see "OPENET," <https://openetdata.org/>.

<sup>64</sup> NASA, "FARMing with Data: OpenET Launches new Tool for Farmers and Ranchers," March 4, 2025, <https://www.nasa.gov/centers-and-facilities/ames/ames-science/farming-with-data-openet-launches-new-tool-for-farmers-and-ranchers/>.

<sup>65</sup> For additional information on the USDA disaster assistance programs, see CRS Report RS21212, *Agricultural Disaster Assistance*, by Christine Whitt; and CRS Report R42854, *Emergency Assistance for Agricultural Land Rehabilitation*, by Megan Stubbs.

<sup>66</sup> Steven Wallander, Elizabeth Marshall, and Marcel Aillery, "Farms Employ Strategies to Reduce Risk of Drought Damages," USDA, Economic Research Service, *Amber Waves*, June 5, 2017, <https://www.ers.usda.gov/amber-waves/2017/june/farmers-employ-strategies-to-reduce-risk-of-drought-damages/>. Hereinafter, Wallander, Marshall, and Aillery, "Drought Damages."

<sup>67</sup> Wallander, Marshall, and Aillery, "Drought Damages."

<sup>68</sup> For additional information on the USDA disaster assistance programs, see CRS Report RS21212, *Agricultural Disaster Assistance*, by Christine Whitt; and CRS Report R42854, *Emergency Assistance for Agricultural Land Rehabilitation*, by Megan Stubbs.

receive mandatory funding amounts of “such sums as necessary.”<sup>69</sup> USDA’s Farm Service Agency (FSA) administers the following direct payment programs, for which producers may file applications through local FSA offices.<sup>70</sup>

- **Livestock Forage Disaster Program (LFP).** LFP provides payments to eligible livestock producers who suffered a loss of grazing forage for eligible livestock due to drought on privately owned or cash-leased pastureland (including cropland planted specifically for grazing).<sup>71</sup> A county’s U.S. Drought Monitor intensity level and the drought’s severity and duration trigger payment. LFP payments for drought are equal to 60% of the monthly feed cost for all covered livestock for up to five months, depending on the drought’s severity. For producers who sold livestock because of drought conditions, the payment rate is equal to 80% of the estimated monthly feed cost.<sup>72</sup>
- **Livestock Indemnity Program (LIP).** LIP provides payments to eligible livestock owners and contract growers for livestock deaths in excess of normal mortality caused by adverse weather. Drought is only an eligible adverse weather event when associated with *Mycoplasma bovis* (M. bovis) in bison and *Bacillus anthracis* (anthrax), a disease that may occur because of drought and directly results in the death of eligible livestock.<sup>73</sup>
- **Emergency Assistance for Livestock, Honeybees, and Farm-Raised Fish Program (ELAP).** ELAP provides payments to producers of livestock, honeybees, and farm-raised fish as compensation for losses due to disease, adverse weather, feed or water shortages, or other conditions not covered under LIP or LFP. ELAP may cover the cost of transporting water and feed but not the cost of water and feed. ELAP also may cover the cost of hauling livestock to forage or other grazing acres during periods of drought.
- **Tree Assistance Program (TAP).** TAP makes payments to qualifying orchardists and nursery tree growers to replant or rehabilitate trees, bushes, and vines damaged by natural disasters, including excessive wind and qualifying drought. Insurance programs generally cover losses in crop production (see “Insurance,” below).

---

<sup>69</sup> The four direct-payment programs are authorized under Section 1501 of the Agricultural Act of 2014, as amended (P.L. 113-79; 7 U.S.C. §9081). For additional information on the four programs, see CRS Report RS21212, *Agricultural Disaster Assistance*, by Christine Whitt.

<sup>70</sup> To find a local Farm Service Agency (FSA) office, see <https://www.farmers.gov/working-with-us/service-center-locator>.

<sup>71</sup> Fire is also an eligible cause of loss on rangeland managed by a federal agency.

<sup>72</sup> For more information about the USDA’s Livestock Forage Disaster Program, see CRS Report R48082, *Livestock Forage Disaster Program (LFP): Drought and Wildfire Assistance*, by Christine Whitt, and CRS Report R48574, *One Big Beautiful Bill Act (H.R. 1): Title I, Farm Safety Net and Miscellaneous Provisions*, coordinated by Stephanie Rosch.

<sup>73</sup> Anthrax is caused by *Bacillus anthracis*, a spore-forming bacterium that can survive in the soil for decades. Hoofed animals (e.g., cattle, goats, sheep) are the main animals affected by this disease. Outbreaks usually occur after periods of drought followed by heavy rains. Spores can become concentrated on the soil surface and on vegetation, where foraging animals can become exposed to the disease. For additional information, see World Organization for Animal Health (WOAH), “Anthrax,” <https://www.woah.org/en/disease/anthrax/>. *Mycoplasma bovis* (M. bovis) is a bacterial pathogen. M. bovis is often transmitted from bison to bison through instances of adverse weather conditions that cause stressors for bison and enhance transmission and symptoms. Bison often carry M. bovis, sometimes without transmission, until it is triggered by stressors, including drought, extreme cold, and oscillating temperature that includes precipitation. USDA, APHIS Veterinary Services, “Mycoplasma Bovis – An Emerging Pathogen in Ranches Bison”, *Emerging Disease Notice*, October 2013, [https://www.aphis.usda.gov/sites/default/files/bison14\\_mbavis.pdf](https://www.aphis.usda.gov/sites/default/files/bison14_mbavis.pdf).

### U.S. Drought Monitor and USDA Programs

The U.S. Drought Monitor is a map that identifies areas of drought and drought intensity using four levels of drought intensity (see “Drought Classification”). Congressional interest in the U.S. Drought Monitor has grown as additional U.S. Department of Agriculture (USDA) program benefits have become tied to its drought classifications. Below are examples of how both Congress and USDA have increasingly relied on the U.S. Drought Monitor as a mechanism for triggering drought assistance in USDA programs.

- Beginning in 2003, USDA used the U.S. Drought Monitor to determine state and county eligibility to distribute nonfat dry milk surplus stocks for livestock feed in areas experiencing extreme or exceptional drought (D3 or D4 classifications).
- In the Food, Conservation, and Energy Act of 2008 (2008 farm bill; P.L. 110-246), Congress required that specific drought intensity classifications published in the U.S. Drought Monitor be used as triggers for payments under the Livestock Forage Disaster Program (LFP). Subsequent reauthorizations of LFP have retained this funding mechanism and use of the U.S. Drought Monitor.
- In 2012, USDA amended the secretarial disaster designation process used to trigger emergency farm loans (see “Loans”) by creating an expedited process—referred to as a *fast-track designation*—for severe drought situations. This fast-track process is based on drought intensity levels from the U.S. Drought Monitor. The process has allowed USDA to issue secretarial disaster designations quickly during periods of severe drought (D2 or higher), expediting access to aid and reducing reporting requirements at the Farm Service Agency (FSA) office level.
- The Agriculture Improvement Act of 2018 (2018 farm bill; P.L. 115-334) amended the Conservation Reserve Program (CRP; see “Conservation”) to allow emergency haying and grazing on selected CRP acres when a county is experiencing drought classified as severe (D2) or greater.
- In the FY2020 Further Consolidated Appropriations Act (P.L. 116-94), Congress repurposed funding for the Wildfires and Hurricanes Indemnity Program (WHIP) to cover losses related to drought, among other loss types. Drought-related losses must have occurred in counties with a D3 (extreme) or D4 (exceptional) classification in calendar years 2018 and 2019.
- The Extending Government Funding and Delivering Emergency Assistance Act (P.L. 117-43, Division B) included \$10 billion for agricultural producers negatively affected by natural disasters, including drought. Congress required that drought-related losses must have occurred in counties with a D2 classification for eight consecutive weeks, or a D3 or higher classification in calendar years 2020 and 2021. The same requirement was present for \$3.74 billion appropriated in the Disaster Relief Supplemental Appropriations Act, 2023 (passed as Division N of the Consolidated Appropriations Act, 2023 (P.L. 117-328)) for drought-related losses occurring in calendar year 2022. The same requirement was also present for \$2 billion appropriated in the Disaster Relief Supplemental Appropriations Act, 2025 (passed as Division B of the American Relief Act, 2025 (P.L. 118-158)) for drought-related losses occurring in 2023 and 2024.

**Sources:** USDA, FSA, “Sale of Surplus Non-fat Dry Milk,” fact sheet, April 2003, [https://www.fsa.usda.gov/Internet/FSA\\_File/nfdm03.pdf](https://www.fsa.usda.gov/Internet/FSA_File/nfdm03.pdf); USDA, FSA, “Disaster Designation Process,” 77 *Federal Register* 41248, July 13, 2012. For further information regarding WHIP, see CRS In Focus IFI 1539, *Wildfires and Hurricanes Indemnity Program (WHIP)*, by Megan Stubbs.

## Insurance

USDA administers two insurance programs that offer subsidized or federally supported insurance coverage for yield, revenue, or other financial losses associated with production of eligible crops and livestock. Coverage is available for adverse weather conditions, including natural disasters, and, in some cases, market declines. Most policies consider drought and related conditions, such as extreme heat and irrigation water supply failure, eligible causes of losses. Coverage is available for *catastrophic* losses—losses in excess of 50% of normal yield. Higher coverage levels may be purchased for less severe losses (referred to as *buy-up* coverage).<sup>74</sup> Policies must be

<sup>74</sup> Buy-up coverage is available in increments of 5% through the Federal Crop Insurance Program (FCIP) to cover between 50% and 85% of a crop and through the Noninsured Crop Disaster Assistance Program (NAP) to cover between 50% and 65% of a crop. For example, a NAP policy with buy-up coverage of 60% would insure losses greater than 40% of the expected yield and provide no coverage for losses amounting to less than 40% of expected yield.

purchased prior to a disaster event, and producers must purchase or renew coverage on an annual basis. These programs are permanently authorized and have mandatory funding authority.

- **Federal Crop Insurance Program (FCIP).** FCIP offers farmers the opportunity to purchase insurance coverage against financial losses caused by various perils, including certain adverse growing and market conditions.<sup>75</sup> Crop insurance is available for most major crops, many specialty crops (e.g., fruit, tree nut, vegetable, and nursery crops), forage and pastureland for livestock producers, and revenues from dairy and livestock production. USDA's Risk Management Agency administers the program, and approved private insurance companies sell and service federal crop insurance policies. Producers must contact their crop insurance agents to file a claim following a loss.<sup>76</sup>
- **Noninsured Crop Disaster Assistance Program (NAP).** NAP provides coverage for crops and in locations where FCIP insurance is unavailable with limited exceptions.<sup>77</sup> NAP applicants must pay an administrative fee at the time of application, plus any additional cost for buy-up coverage. FSA administers NAP, and producers must notify their local FSA office following a loss.

## Cost-Share Assistance

Some USDA programs pay a percentage of the cost to reinstall conservation infrastructure or rehabilitate damaged land. Advance sign-up generally is not required. However, the programs will not pay for impairments existing before the disaster event. FSA administers the following cost-share programs, for which producers may file applications through local FSA offices. These programs are permanently authorized but subject to appropriations.<sup>78</sup>

- **Emergency Conservation Program (ECP).** ECP assists landowners in restoring land used in agricultural production damaged by a natural disaster and in implementing emergency water-conservation measures in severe drought periods. Eligible activities may include providing emergency water for livestock and existing permanently installed irrigation systems for orchards and vineyards.
- **Emergency Forest Restoration Program (EFRP).** EFRP provides cost-share assistance to private forestland owners to repair and rehabilitate damage caused by natural disasters, including drought, on nonindustrial private forestland. Eligible practices may include removing debris and replanting to restore forest-related damage from drought.

## Loans

USDA can provide low-interest loans to help producers recover from production or physical losses due to drought and other natural disasters, or it can provide temporary loan relief for

---

<sup>75</sup> FCIP is authorized by the Federal Crop Insurance Act, as amended (P.L. 96-365; 7 U.S.C. §§1501 et seq.). For more information on FCIP, contact Stephanie Rosch, Analyst in Agriculture Policy, or see CRS Report R46686, *Federal Crop Insurance: A Primer*, by Stephanie Rosch.

<sup>76</sup> To locate an approved insurance provider, visit <https://public-rma.fpac.usda.gov/AipListing/InsuranceProviders>.

<sup>77</sup> NAP is authorized under Section 196 of the Federal Agriculture Improvement and Reform Act of 1996, as amended (P.L. 104-127; 7 U.S.C. §7333). For additional information on NAP, see CRS Report R48245, *The Noninsured Crop Disaster Assistance Program (NAP)*, by Christine Whitt.

<sup>78</sup> The cost-share assistance programs are authorized under Title IV of the Agricultural Credit Act of 1978 (P.L. 95-334; 16 U.S.C. §§2201-2206). For additional information on the programs, see CRS Report R42854, *Emergency Assistance for Agricultural Land Rehabilitation*, by Megan Stubbs.

existing FSA farm loans. These loan options are triggered when a county is designated as a disaster area under a presidential major disaster declaration, a presidential emergency declaration pursuant to the Robert T. Stafford Disaster Relief and Emergency Assistance Act (Stafford Act; 42 U.S.C. §§5121 et seq.), or a USDA secretarial disaster designation. Stafford Act declarations related to drought are uncommon. USDA may issue a secretarial disaster designation quickly during periods of severe drought using a fast-track process in accordance with U.S. Drought Monitor intensity (see text box on “U.S. Drought Monitor and USDA Programs,” above). FSA administers the loan programs, and local FSA offices accept applications. USDA loan programs are subject to appropriations.<sup>79</sup>

- **Emergency Farm Loans.** Loans may help producers recover from production and physical losses.<sup>80</sup> A qualified applicant can borrow up to 100% of actual production (e.g., loss of a crop) or physical losses (e.g., repairing or replacing damaged or destroyed structures or replanting permanent crops, such as orchards). Loan totals may not exceed \$500,000. A producer must operate in a county declared eligible or in a contiguous county and must meet the loan requirements.
- **Disaster Set-Aside.** A producer with an existing FSA farm ownership or operating loan located in a designated disaster area or contiguous county may apply to set aside one payment to allow the operation to continue.<sup>81</sup> The payment set-aside is not forgiven and must be repaid prior to final maturity of the note. Any principal set-aside continues to accrue interest until repaid.

## Conservation

Certain USDA conservation programs that are not emergency programs may provide assistance during drought periods to help alleviate the drought’s impacts on agricultural production.<sup>82</sup> In many cases, this assistance comes through the Secretary of Agriculture’s use of waivers and authority to provide flexibility in program implementation. Other assistance is offered through adaptive measures that reduce drought risk through various management decisions and practices, which are not discussed in detail in this report.<sup>83</sup>

---

<sup>79</sup> For more information on USDA loan programs, see CRS Report R46768, *Agricultural Credit: Institutions and Issues*, by Jim Monke.

<sup>80</sup> Emergency farm loans are authorized under Title III of the Consolidated Farm and Rural Development Act, as amended (7 U.S.C. §§1961 et seq.). For more information, see USDA, FSA, “Emergency Farm Loans,” <https://www.fsa.usda.gov/resources/loans/emergency-farm-loans>. Interest rates are available <https://www.fsa.usda.gov/tools/informational/rates/current-fsa-loan-interest-rates>.

<sup>81</sup> The Disaster Set-Aside Program is authorized under Section 331A of the Consolidated Farm and Rural Development Act, as amended (7 U.S.C. §1981a). For more information, see, USDA, FSA, “Disaster Set-Aside Program,” fact sheet, May 2024, <https://www.fsa.usda.gov/tools/informational/fact-sheets/disaster-set-aside-program-2024pdf>. See also the Distressed Borrower Set Aside program that does not require a disaster designation (USDA, “Distressed Borrower Set-Aside Program,” fact sheet, August 2024, [https://www.fsa.usda.gov/sites/default/files/documents/fact\\_sheet-distressed\\_borrower\\_set\\_aside\\_program.pdf](https://www.fsa.usda.gov/sites/default/files/documents/fact_sheet-distressed_borrower_set_aside_program.pdf)).

<sup>82</sup> For additional information on USDA conservation programs generally, see CRS In Focus IF12024, *Farm Bill Primer: Conservation Title*, by Megan Stubbs.

<sup>83</sup> Examples of *adaptive measures* include soil health practices—such as no till or reduced tillage, cover crop adoption, and conservation crop rotations—that can increase soil organic matter over time and reduce soil moisture loss. Adoption of irrigation efficiency practices—such as variable-rate irrigation, flow meters, land leveling, and soil moisture sensors—generally reduces evaporation and runoff loss but also may affect ground water infiltration and downstream water availability.

- **Conservation Reserve Program (CRP).** CRP uses mandatory funding to provide annual payments to agricultural producers to take highly erodible and environmentally sensitive land out of production and install resource-conserving practices for 10 or more years.<sup>84</sup> Haying and grazing may occur on CRP acres under emergency and certain nonemergency conditions. Emergency haying and grazing of CRP acres is generally authorized during periods of severe drought (D2 or higher, according to the U.S. Drought Monitor) or similar natural disaster (e.g., wildfire).<sup>85</sup> Outside of the primary nesting season, up to 90 days of grazing or one cutting of hay is allowed.<sup>86</sup> During the primary nesting season, emergency grazing is allowed in approved counties, but at half the normal carrying capacity and only if LFP payments also have been triggered for the county. Emergency haying is not allowed on CRP acres during the primary nesting season. Not all CRP practices are eligible for haying and grazing; a request must be filed with the local FSA office before any activity begins.
- **Environmental Quality Incentives Program (EQIP).** EQIP is a voluntary program that uses mandatory funding to provide financial and technical assistance to agricultural producers to address natural resource concerns on agricultural and forestland.<sup>87</sup> In the past, USDA has announced special EQIP sign-ups for farmers and ranchers in hurricane-, flood-, or drought-affected areas. EQIP also may be used to proactively mitigate potential damage from natural disasters through conservation practices (e.g., residue management to improve the soil's capacity to be more drought resilient). NRCS administers EQIP, and applications may be filed at any local NRCS office.<sup>88</sup>

## NOAA Drought Support Programs

Congress has furnished the Secretary of Commerce, through NOAA, with the authority to provide disaster assistance to the fishing industry when fish populations decline or other disruptions cause economic loss. The Secretary has declared fishery resource disasters specifically related to drought in several cases, most recently for Florida oysters in 2012.<sup>89</sup> In 2022, Congress amended

---

<sup>84</sup> The Conservation Reserve Program (CRP) is authorized under Title XII of the Food Security Act of 1985, as amended (P.L. 99-198; 16 U.S.C. §§3831-3835). For additional information on CRP emergency haying and grazing, see USDA, FSA, "CRP Haying and Grazing: Non-Emergency and Emergency Use," fact sheet, August 2025, <https://www.usda.gov/guidance-documents/disaster-protection-and-recovery/fsa/emergency-conservation-reserve-program-crp-haying-and-grazing>.

<sup>85</sup> Emergency haying and grazing status is reviewed by FSA every Thursday using the U.S. Drought Monitor. Approved counties are listed on the USDA, FSA "Counties Approved for Emergency Haying and/or Grazing" website, <https://www.fsa.usda.gov/resources/programs/conservation-reserve-program/emergency-haying-grazing>.

<sup>86</sup> The primary nesting season is established in each state as the nesting season for birds in the local area that are economically significant, in significant decline, or conserved in accordance with federal or state law (7 C.F.R. §1410.2). For a list of primary nesting dates and durations, see USDA, FSA, "Primary Nesting Season Dates and Duration," June 16, 2020, <https://www.fsa.usda.gov/documents/primary-nesting-season-june-16-2020pdf>.

<sup>87</sup> The Environmental Quality Incentives Program (EQIP) is authorized under Title XII of the Food Security Act of 1985, as amended (P.L. 99-198; 16 U.S.C. §§3839aa-3839aa-7). For additional information on EQIP, see USDA, NRCS, "Environmental Quality Incentives Program," <https://www.nrcs.usda.gov/wps/portal/nrcs/main/national/programs/financial/eqip/>.

<sup>88</sup> To find a local NRCS office, see <https://www.farmers.gov/working-with-us/service-center-locator>.

<sup>89</sup> NOAA, "Fishery Disaster Determinations," <https://www.fisheries.noaa.gov/national/funding-and-financial-services/fishery-disaster-determinations>. Prior to 2022, Congress provided multiple authorities to the Secretary of Commerce to respond to fishery disasters, including due to "natural causes." For more information about these provisions, see CRS (continued...)

the Secretary's authority, explicitly listing drought, among other phenomena, as a *natural cause* that may be considered under a fishery resource disaster determination.<sup>90</sup> Once a review is initiated and necessary information is obtained and reviewed, the Secretary determines whether a fishery failure or *fishery resource disaster* has occurred.<sup>91</sup> In most cases, Congress has appropriated funds to support the fishing industry following the Secretary's determination. However, since FY2023, Congress intermittently has provided funding to NOAA's National Marine Fisheries Service (NMFS) for anticipated fishery resource disasters.<sup>92</sup> NMFS, states, regional commissions, and industry representatives often work together to plan how to distribute assistance to the fishing industry and allocate assistance among potential projects.

## Drought and Federal Water Resource Facilities and Projects<sup>93</sup>

Federal water resource facilities address or respond to drought in various ways based on different authorities that underpin their responsibilities and activities. Individual authorities or operational directions may provide for a variety of drought-response actions, depending on the agency and location. The federal government owns and operates thousands of dams and other infrastructure that support water supplies. These facilities' operations, particularly the water stored in reservoirs at federal dams, can both assist in meeting water supply needs during droughts and be vulnerable to droughts. Federal dams, particularly in the West, were constructed in part to provide multiyear storage to help with variations in seasonal and annual precipitation.

Most large-scale federal water resource projects were built and are managed by the two principal federal water resource agencies: Reclamation and USACE. The discussion herein focuses on the federal dams that form storage reservoirs operated by these two federal agencies. **Figure 7** shows the Reclamation and USACE reservoirs with capacities greater than 25,000 acre-feet (AF) that had water supply or irrigation among their operational purposes as of 2021.<sup>94</sup> Federal reservoirs are distributed across the coterminous United States; in some watersheds and states, particularly in the West, federal reservoirs and related infrastructure are prevalent and play a significant role in storing and delivering water supplies.

---

Report RL34209, *Fishery Disaster Assistance*, by Anthony R. Marshak. Congress amended or repealed these authorities in the Fishery Resource Disasters Improvement Act, Division S, Title II of P.L. 117-328, and the Fishery Improvement to Streamline untimely regulatory Hurdles post Emergency Situation Act (FISHES Act; P.L. 118-229).

<sup>90</sup> Fishery Resource Disasters Improvement Act, Division S, Title II of P.L. 117-328. 16 U.S.C. §1861a(a)(1)(E)(VI).

<sup>91</sup> The Magnuson-Stevens Fishery Conservation and Management Act (MSA), under 16 U.S.C. §1861a(a)(1)(C), defines a *fishery resource disaster* as “a disaster that is determined by the Secretary [of Commerce] in accordance with this subsection [i.e., 16 U.S.C. §1861a(a)] and—(i) is an unexpected large decrease in fish stock biomass or other change that results in significant loss of access to the fishery resource, which may include loss of fishing vessels and gear for a substantial period of time and results in significant revenue loss or negative subsistence impact due to an allowable cause; and (ii) does not include—(I) reasonably predictable, foreseeable, and recurrent fishery cyclical variations in species distribution or stock abundance; or (II) reductions in fishing opportunities resulting from conservation and management measures taken pursuant to this chapter.”

<sup>92</sup> Congress appropriated \$300 million to remain available until expended “for necessary expenses associated with the mitigation of fishery disasters” in the Consolidated Appropriations Act, 2023 (P.L. 117-328, Division N, Title II). Similarly, Congress appropriated another \$300 million to remain available until expended for fishery resource disaster relief in the Disaster Relief Supplemental Appropriations Act, 2025 (P.L. 118-158, Division B, Title II).

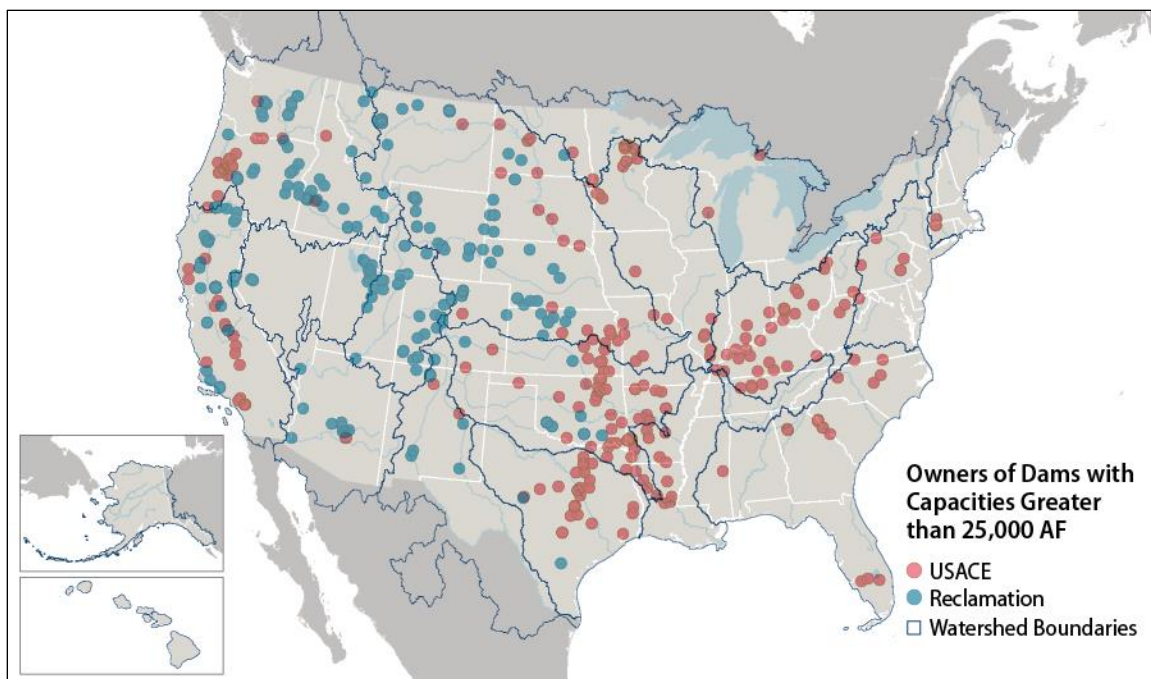
<sup>93</sup> For more information on the Bureau of Reclamation, contact Charles V. Stern, Specialist in Natural Resources Policy. For more information on USACE, contact Nicole T. Carter, Specialist in Natural Resources Policy. For more information on the Endangered Species Act, contact Pervaze A. Sheikh, Specialist in Natural Resources Policy.

<sup>94</sup> An acre-foot of water is equal to the volume of a sheet of water 1 acre in area and 1 foot in depth, equivalent to 43,560 cubic feet of water.

Reclamation is a central player in water resource management in the West. Reclamation maintains 491 dams and other water supply infrastructure in the 17 arid and semi-arid western states (known as *reclamation states*), as defined in statute.<sup>95</sup> Reclamation's facilities provide water supplies to more than 31 million people in the West and deliver a total of nearly 30 million AF of water annually. During droughts, these facilities face operational challenges and particular scrutiny, in part due to conflicting priorities among users of the water that Reclamation facilities supply.

**Figure 7. Bureau of Reclamation and U.S. Army Corps of Engineers Dams with Water Supply and/or Irrigation Purposes**

(dams with storage capacities greater than 25,000 acre-feet as of 2021)



**Source:** CRS, using data from the National Inventory of Dams (NID) and U.S. Geological Survey watershed boundaries, data accessed in August 2021.

**Notes:** AF = Acre-feet (1 AF of water is equal to the volume of a sheet of water 1 acre in area and 1 foot in depth, equivalent to 43,560 cubic feet of water); Reclamation = U.S. Bureau of Reclamation; and USACE = U.S. Army Corps of Engineers. Municipal and industrial water supply and irrigation water storage generally are not primary purposes of USACE projects and instead are associated with multipurpose USACE projects. This figure refers to Reclamation and USACE as the dam owners for consistency with NID classifications; these dams are officially assets of the federal government administered by the agencies.

<sup>95</sup> An 1890-1896 drought coincided with a period in U.S. history of federal encouragement of large-scale efforts to irrigate the relatively arid western states. At that time, Congress debated a larger federal role in western states' irrigation. This debate led to the Reclamation Act of 1902, which was enacted largely to "reclaim the arid West." The federal government constructed hundreds of dams, reservoirs, and related facilities to provide water to local farmers to reclaim the arid West through irrigation of arid lands. Today, Reclamation is responsible for managing and developing many of the large federal dams and water diversion structures in the 17 coterminous states west of the Mississippi River (referred to as *reclamation states*): Arizona, California, Colorado, Idaho, Kansas, Montana, Nebraska, Nevada, New Mexico, North Dakota, Oklahoma, Oregon, South Dakota, Texas, Utah, Washington, and Wyoming. For more information, see CRS Report R46303, *Bureau of Reclamation: History, Authorities, and Issues for Congress*, by Charles V. Stern and Anna E. Normand.

In contrast, USACE facilities operate throughout the United States and typically have not focused on water supply storage as a principal project purpose; instead, water supply storage is often a secondary purpose of multipurpose USACE facilities. The role of USACE in water storage nonetheless can be regionally or locally significant. As of FY2023, USACE had agreements to provide municipal and industrial (M&I) water supply storage totaling 10.4 million AF at 135 projects, benefitting hundreds of communities.<sup>96</sup>

Sustained hydrological drought has operational effects on federal reservoirs, dams, locks, hydroelectric facilities, and other components of the nation's water infrastructure. For example, numerous USACE reservoirs have drought management plans that result in the curtailing of some benefits (e.g., navigation, hydropower) to maintain other benefits (e.g., in-stream flows to support water quality, aquatic species, and river withdrawals for electric power cooling and municipal and industrial water supplies).

Federal facilities' ability to deliver water supplies can be curtailed by low inflow and storage conditions accompanying droughts. In the past, Congress has conducted oversight of federal facilities' ongoing operations and has been active in several other areas related to federal reservoirs and drought, including the following activities discussed in more detail below:

- Providing authorities and funding to expand western water storage;
- Enacting authorities related to updating operating plans for reservoirs during drought;
- Supporting developments in forecast-informed reservoir operations (FIRO) and other efforts to enhance the data used to inform operations decisions; and
- Allowing for regulatory relief, in some instances, for federal water resource project operations during drought

## **New and Expanded Water Storage Projects**

### ***Reclamation***

Some long-term federal drought response authorities support the construction of new or expanded water storage projects in drought-prone areas. In 2015, Congress created a new authority for Reclamation to build water storage: Section 4007 of the Water Infrastructure Improvements for the Nation Act (WIIN Act; P.L. 114-322). The authority provided federal support for the construction of new and augmented surface and groundwater storage projects in the 17 reclamation states.

Funding for water storage projects under Section 4007 is available for two primary project types. Federally owned storage projects may receive federal funding for up to 50% of project costs, with project sponsors responsible for the remainder of costs. State-led storage projects may receive up to 25% of their costs as federal funding. Prior to the WIIN Act, Congress had not authorized Reclamation to fund state-led water storage projects.

Before the federal government can spend or contribute funding for the construction of new water projects under this authority, several criteria must be met:

---

<sup>96</sup> USACE Institute for Water Resources, *FY 2023 Municipal, Industrial, and Irrigation Water Supply Database Report*, 2025-WRC-03, May 2025.

- The Secretary of the Interior must find that the project is feasible and provides benefits proportionate to the federal government's cost share, and project sponsors must agree to pay their portions of project costs up front.
- The Secretary of the Interior must transmit a list of recommended projects and funding levels to Congress, and Congress must designate those projects by name in an enacted appropriations act.

The WIIN Act set a deadline of January 1, 2021, for projects to be found feasible by Reclamation and be eligible to receive additional construction funding. Congress approved Reclamation study funding prior to that date for 13 projects in three states (California, Idaho, and Washington). Eight of these projects were found feasible prior to the deadline and thus are eligible for ongoing funding.<sup>97</sup> Congress has subsequently funded these projects in regular and supplemental appropriations legislation.

## USACE

Water supply and related storage historically were not principal project purposes of USACE-constructed facilities in the United States; however, since 2020, Congress has expanded the agency's water supply- and drought-related authorities (see **Table 1**). Actions on these USACE authorities typically are subject to the availability of appropriations. Congress also included a policy declaration in the Grace F. Napolitano Priority for Water Supply, Water Conservation, and Drought Resiliency Act of 2024, enacted as Title I, Subtitle B, of Division A of P.L. 118-272, the Thomas R. Carper Water Resources Development Act of 2024 (WRDA 2024). It states that it is U.S. policy for USACE to maximize opportunities for water supply and conservation measures and drought resilience efforts at and in the operation of the agency's water resource projects.

**Table 1. Selected USACE Water Supply and Drought Authorities Enacted Since 2020**

| Topic                                                      | USACE Authority Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Water Supply Storage Projects                              | In 2020 (P.L. 116-260), Congress provided USACE with a new general authority to carry out certain water storage projects, including for water supply and water conservation. The authority allowed USACE to undertake construction of (1) new water storage projects with capacities between 2,000 AF and 30,000 AF or (2) enlargements between 1,000 AF and 30,000 AF of existing storage. It limits federal project costs to \$65 million and specifies that municipal and industrial costs are 100% nonfederal and agricultural water supply cost is 35% nonfederal. <sup>a</sup> |
| Water Supply and Water Conservation in Feasibility Studies | Congress in 2022 (P.L. 117-263) provided USACE with additional authorities to study water supply and water conservation activities as part of its feasibility studies for new or modified water resources projects.<br>Congress, in legislation passed in 2024 (and signed in early 2025, P.L. 118-272), authorized USACE to conduct selected single-purpose water supply feasibility studies.                                                                                                                                                                                       |
| Aquifer Recharge Assessment                                | Congress in 2022 (P.L. 117-263) authorized a national assessment of using managed aquifer recharge projects at authorized water resource development projects to address drought, reduce aquifer depletion and enhance water resilience, and to assess and identify opportunities to support nonfederal interests in carrying out managed recharge projects.                                                                                                                                                                                                                         |
| Aquifer Recharge Working Group                             | Congress in 2022 (P.L. 117-263) directed the establishment of a USACE-nonfederal aquifer recharge working group.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

<sup>97</sup> For more information, see CRS In Focus IF10626, *Reclamation Water Storage Projects: Section 4007 of the Water Infrastructure Improvements for the Nation Act*, by Charles V. Stern.

| Topic                                | USACE Authority Description                                                                                                                                                                                                      |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aquifer Recharge Feasibility Studies | Congress in 2022 (P.L. 117-263) authorized USACE to perform up to 10 feasibility studies (at 90% federal cost and 10% nonfederal cost) on managed aquifer recharge projects in drought-prone or water-scarce areas. <sup>b</sup> |
| Drought Measures                     | Congress in P.L. 118-272 incorporated drought measures as activities into some of USACE's existing programmatic authorities for projects with federal costs less than \$15 million. <sup>c</sup>                                 |

**Source:** CRS using referenced legislation.

- a. The Assistant Secretary of the Army for Civil Works (ASACW) developed implementation guidance for the program (Memorandum from Michael L. O'Connor, ASACW, to Commanding General, USACE, "Implementation Guidance for Section 155 of the Water Resources Development Act of 2020, Small Water Storage Projects," April 25, 2022, <https://usace.contentdm.oclc.org/utis/getfile/collection/p16021coll5/id/35987>).
- b. The ASACW's implementation guidance indicates that the managed aquifer recharge portion of a study undertaken pursuant to this authority uses the 10% nonfederal cost share and that construction costs follow existing USACE cost sharing for water supply set out at 33 U.S.C. §2213, which is 100% nonfederal for M&I water supply and 35% nonfederal for agricultural water supply (Assistant Secretary of the Army (Civil Works), Memorandum from Jaime A. Pinkham, Acting ASACW, to Commanding General, USACE, "Implementation Guidance for Section 8108(b) of the Water Resources Development Act of 2022, Managed Aquifer Recharge Feasibility Studies," January 17, 2025, <https://usace.contentdm.oclc.org/utis/getfile/collection/p16021coll5/id/38410>).
- c. For more on these programmatic authorities, see CRS In Focus IF12635, *Continuing Authorities Programs (CAPs) of the U.S. Army Corps of Engineers*, by Nicole T. Carter and Anna E. Normand.

## Drought Operations Manuals and Planning

### Reclamation

Reclamation projects typically are governed by operating plans.<sup>98</sup> In Section 202 of the Reclamation States Emergency Drought Relief Act of 1991 (P.L. 102-250), Congress authorized Reclamation to prepare or participate in the preparation of cooperative drought contingency plans, for the prevention or mitigation of adverse effects of drought conditions.<sup>99</sup> There is no statutory requirement for Reclamation to formally revise its operating plans with drought contingency plans (DCPs). In practice, Reclamation generally has supported contingency plans developed by nonfederal interests through its Drought Response Program. Most Reclamation contracts with water users include provisions that allow Reclamation to restrict deliveries due to water shortages and other drought-related factors, which allows for some level of operational flexibility during drought.<sup>100</sup>

In part to facilitate Reclamation's ability to adapt its operations to changes in water supplies and demand, in 2009 Congress enacted Section 9503(c) of the SECURE Water Act (P.L. 111-11). This directed Reclamation to identify and assess potential water-related risks of climate change in major reclamation river basins and to report on these efforts at five-year intervals.<sup>101</sup> Reclamation published reports pursuant to the SECURE Water Act in 2011, 2016, and 2021.<sup>102</sup> These reports,

<sup>98</sup> In accordance with Section 7 of the Flood Control Act of 1944 (33 U.S.C. §709), USACE has formal responsibility for preparing water-control manuals for reclamation projects with storage space allocated for flood control purposes.

<sup>99</sup> 43 U.S.C. §2222.

<sup>100</sup> See Bureau of Reclamation, "Water-Related Contracts and Charges – General Principles and Requirements," Reclamation Manual PEC P05, July 24, 2013, <https://www.usbr.gov/recman/pec/pec-p05.pdf>.

<sup>101</sup> 42 U.S.C. §10363.

<sup>102</sup> For more information and individual reports, see <https://www.usbr.gov/climate/secure/2021secure.html>.

coupled with complementary reporting in the form of West-wide climate and hydrology assessments and individual river basin reports, highlight Reclamation's analysis of and expectations for altered water supply and demand at a variety of geographical levels. Although drought is not the primary purpose of these reports, they contribute toward bureau efforts to assess and address droughts.

The SECURE Water Act also authorized Reclamation to undertake actions to address climate change risks. The act authorized Reclamation to use specific strategies for this purpose, including ones related to (1) modifying reservoir storage or operating guidelines; (2) developing new water management, operating, or habitat restoration plans; (3) water conservation; (4) improved hydrologic models and other decision support systems; and (5) groundwater and surface water storage needs.<sup>103</sup> One way Reclamation has used these authorities is by conducting a number of reservoir operations pilot studies at specific locations in the 17 reclamation states; these studies seek to identify operational improvements to respond to variable water supplies, including droughts. Beginning in 2015, Reclamation conducted five pilot studies.<sup>104</sup> In June 2021, Reclamation announced another six pilot studies in five western states.<sup>105</sup> Initial studies employed a variety of approaches, including the increased use of pre-historic hydrology, advanced monitoring, and forecast-informed operations, among others (for more information on forecast-informed operations, see "Forecast-Informed Reservoir Operations" below).

## **USACE**

USACE maintains water-control manuals that detail how reservoirs are to be operated given their congressionally authorized purposes. Since the late 1970s, USACE regulations have required USACE projects that maintain controlled reservoir storage to have a DCP. The DCP informs water management decisions and responses to drought-related water shortages in a basin. Because of uncertainties such as when a drought will end, DCPs generally specify a minimum suite of actions related to water control and allow for additional actions as the specific situation warrants (e.g., approval of deviations from operating plans).<sup>106</sup>

In 2015, after an internal assessment of the state of USACE drought contingency planning and an effort to develop methods to update DCPs to account for a changing climate, USACE released a report titled *USACE Drought Contingency Planning in the Context of Climate Change*.<sup>107</sup> The report reviewed the 142 existing DCPs for USACE projects, the majority of which were developed prior to 1993, and noted that "none of the DCPs reviewed include information about drought projections under future climate change. Consequently, it is unlikely that these reports provide an adequate guide for preparing for future droughts that may be longer and more intense than recognized by these DCPs."

---

<sup>103</sup> 42 U.S.C. §10363.

<sup>104</sup> For more information, see Reclamation, "Reservoir Operations Pilots," <https://www.usbr.gov/watersmart/pilots/index.html>.

<sup>105</sup> For more information, see Reclamation, "FY2021 Reservoir Operations Pilots—Round 1 Selections," <https://www.usbr.gov/watersmart/pilots/docs/FY21-ResOps-R1ProjectDescriptions.pdf>.

<sup>106</sup> A *deviation* is the operation of a USACE project in a manner other than specified in the approved water-control manual or its associated drought contingency plan (DCP). A detailed assessment is required prior to the execution of a deviation.

<sup>107</sup> USACE, *USACE Drought Contingency Planning in the Context of Climate Change*, CWTS Report 15-15, September 2015, <https://usace.contentdm.oclc.org/digital/collection/p266001coll1/id/6727>. The report team identified and assessed 142 DCPs covering 301 projects.

In 2018, USACE updated the 1981 regulation that guided the development and updating of DCPs.<sup>108</sup> The 1981 regulation used the standard engineering practice of preparing DCPs based on observed periods of record for temperature, precipitation, and drought. The 2018 regulation identified actions that “at a minimum” its planners and engineers should undertake to incorporate climate change considerations into DCP development, including the consideration of regional variables.

Congress has provided USACE with various general and project-specific authorities related to reservoir management for drought and water conservation activities since 2014.<sup>109</sup> While these authorities are available, they have been used for few reservoirs to date. For example, under the authority of Section 1116 of the WIIN Act, which authorized USACE to study and perform water conservation measures at USACE reservoirs in certain states that had declared droughts, USACE evaluated and approved conservation measures at Prado Dam, Riverside County, CA, for the purpose of downstream nonfederal groundwater recharge efforts.<sup>110</sup> Section 8107 of WRDA 2022 further expanded this authority. Section 1164 of WRDA 2024 authorized the Secretary of the Army to, under a pilot program, operate the flood control pool of certain reservoirs in Arizona, California, and Nevada for the primary purpose of water supply under various constraints during a drought emergency.

Separately, Section 8208 of WRDA 2022 directed the Secretary of the Army to conduct a study of the effectiveness of carrying out additional measures at USACE reservoirs in the South Pacific Division to increase water supply and mitigate drought and flood, among other purposes.<sup>111</sup> WRDA 2022 also addressed western states’ concerns related to how USACE reservoirs operate vis-à-vis state water rights. In contrast to Reclamation facilities that often have state water rights associated with them, USACE typically does not have water rights associated with its reservoirs. WRDA 2022 directed the Secretary of the Army to establish a Western Water Cooperative Committee to work on identifying opportunities to avoid or minimize conflicts between USACE flood control projects and water rights and water laws in western states.<sup>112</sup>

Water supplies for communities and agriculture receive much attention during droughts, but USACE also may make some drought-related adjustments to meet its navigation mission. For

---

<sup>108</sup> USACE, *Drought Contingency Plans*, ER 1110-2-1941, February 2, 2018.

<sup>109</sup> The general authorities include provisions in the Water Resources Reform and Development Act of 2014 (WRRDA 2014; P.L. 113-121) and the Water Infrastructure Improvements for the Nation Act (WIIN Act; P.L. 114-322). In WRRDA 2014, Section 1045, Congress authorized a USACE assessment of the effects of drought conditions on lakes managed by the Secretary of the Army that are affected by Federal Energy Regulatory Commission-licensed reservoirs. In Section 1046, Congress authorized a USACE assessment of the management practices, priorities, and authorized purposes at USACE reservoirs in arid regions to determine the effects on water supply of periods of drought, among other things. In WIIN Section 1117, Congress authorized USACE, at the request of a governor, to prioritize updating (including for seasonal operations for water conservation and supply) the water-control manuals for USACE operated and maintained water-control structures in certain states with declared droughts. Section 8107 of WRDA 2022 also authorized updating water control manuals for states with a statewide disaster declaration in 2021.

<sup>110</sup> Personal communication from USACE staff to CRS staff, July 8, 2021. According to the Assistant Secretary of the Army, the Prado Dam FIRO pilot found “that an average of 7000 acre-feet (AF) per year of stormwater could be released in a modified manner to allow the Orange County Water District to use the water in its groundwater recharge system and provide additional supply for its customers.” (Assistant Secretary of the Army (Civil Works), Memorandum, “Army Civil Works Supporting Drought Resilience in America’s Communities,” July 28, 2022.)

<sup>111</sup> For a map of the South Pacific Division, see USACE, “Around the South Pacific Division,” <https://www.spd.usace.army.mil/>. The division covers the states of Arizona, California, and New Mexico; most of Colorado, Nevada, and Utah; and portions of Texas, Idaho, Oregon, and Wyoming.

<sup>112</sup> For more on the committee and its activities, see USACE, Western Water Cooperative Committee, <https://www.usace.army.mil/Missions/Civil-Works/Project-Planning/Legislative-Links/WRDA-2022/Western-Water-Cooperative-Committee/>.

example, in 2012 and in 2022, USACE maintained navigation on the Mississippi River and its tributaries (albeit in a narrower and shallower channel than is available in a normal water year) using a combination of measures: dredging of critical areas, removal of rock pinnacles, and releases of reservoir water within authorized purposes.

## Forecast-Informed Reservoir Operations

Federal dam operators have often used runoff measurements, runoff estimates, and other observations (e.g., snowpack or soil moisture) to inform decisions related to storing or releasing water while also following water-control manuals that may prescribe seasonal and other operational targets and rules. Advances in weather forecasting, in particular in detecting and forecasting atmospheric rivers,<sup>113</sup> have increased decisionmakers' interest in identifying opportunities to use forecasts for more dynamic dam operations, and in improving precipitation forecasts beyond two weeks.<sup>114</sup> That is, operations would shift from rules that use a single estimate of runoff to rules for making operational decisions based on ensembles of runoff (or streamflow) forecasts and statistical techniques to simulate conditions and operations. This reservoir management approach, referred to as forecast-informed reservoir operations (FIRO), uses data from watershed monitoring and from weather and water forecasting to inform water management decisions to retain or release water from reservoirs. A July 2022 memorandum by the then-Assistant Secretary of the Army for Civil Works states "FIRO and related initiatives are among the most cost-effective ways to increase water availability in drought-impacted regions. In some cases, water availability may be significantly increased on an annual basis for less than 5% of the cost of new infrastructure investments on a dollar per [acre foot] basis."<sup>115</sup>

Researchers and agencies have applied early FIRO efforts to reservoirs in California, a state that is not only pursuing additional water supplies due to being drought-prone but also receives a significant portion of its annual precipitation from atmospheric rivers. An atmospheric river (AR) consists of a long band of water vapor moving through the atmosphere, typically resulting in heavy precipitation over land (**Figure 8**).<sup>116</sup> Reservoir managers try to store more water supplies to meet demands during dry periods while considering the consequences for how quickly reservoir storage space can be made available to capture floodwaters during wet periods. Managers using FIRO make these operational decisions based in part on forecasts, rather than following more traditional fixed operational rules during different seasons and conditions.

---

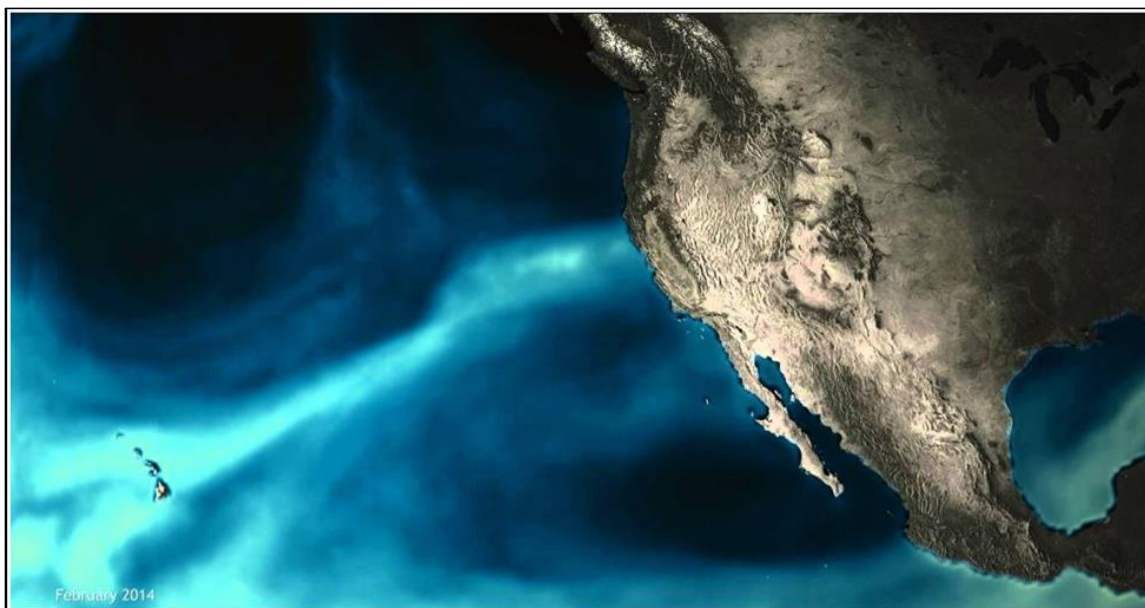
<sup>113</sup> *Atmospheric rivers* are a flowing corridor of concentrated water vapor in the atmosphere that can contribute to significant rainfall or snow upon landfall. For more information, see CRS In Focus IF12872, *Atmospheric Rivers: Background and Forecasting*, by Eva Lipiec and Nicole T. Carter.

<sup>114</sup> In addition to expressing interest in using forecasts, Congress has supported the expansion of the collection and use of observed data in informing reservoir operations and other flood and drought risk preparedness activities in some basins. For example, in 2014 and 2020, Congress authorized actions to improve soil moisture and snowpack monitoring in the Upper Missouri River Basin (P.L. 113-121, §4003(a); P.L. 116-260, §511).

<sup>115</sup> Assistant Secretary of the Army (Civil Works), Memorandum, "Army Civil Works Supporting Drought Resilience in America's Communities," July 28, 2022.

<sup>116</sup> For more on atmospheric rivers, see CRS In Focus IF12872, *Atmospheric Rivers: Background and Forecasting*, by Eva Lipiec and Nicole T. Carter.

**Figure 8. Depiction of Atmospheric River Originating Near Hawaii and Extending to the U.S. West Coast**



**Source:** NOAA, “Atmospheric Rivers,” <https://noaa.maps.arcgis.com/apps/Cascade/index.html?appid=2871beb71aae4542a63b31157c4dce3b>.

**Notes:** In this image from February 2014, the light blue area depicts a plume of water vapor. Atmospheric rivers typically form in tropical regions when winds over the ocean draw water vapor into narrow bands. The atmospheric river’s interactions with land features, such as mountain ranges, or certain atmospheric conditions cause the water vapor to move upward in the atmosphere and fall as heavy rain or snowfall.

Various federal agencies have worked together on a limited number of pilot projects to shift to FIRO-based management, including one applying FIRO at USACE’s Lake Mendocino in California’s Russian River Basin.<sup>117</sup> The Lake Mendocino pilot informed USACE’s 2025 update of the water control manual for the lake and the associated USACE dam that provides flood control and stores water for municipal and agricultural water supplies. The updated manual allows additional water to be stored in Lake Mendocino during the winter rainy season, with the goal of improving drier-season water supply reliability and environmental conditions in the Russian River without harming the reservoir’s flood management function. Reclamation and USACE also incorporated forecasts into a 2018 update to their joint *Folsom Dam Water Control Manual*.<sup>118</sup> Pursuant to the manual, the federal water managers use continuous five-day forecasts available for that region of California to identify plausible flows into the reservoir to assess operational release requirements.

In late 2020, Congress directed the Secretary of the Army to produce a report identifying additional opportunities for applying FIRO across the United States.<sup>119</sup> In July 2026, USACE

<sup>117</sup> Ongoing FIRO research efforts also are underway for Prado Dam, CA; the Yuba-Feather River System, CA; and Howard Hanson Dam, WA. The Lake Mendocino FIRO research effort featured involvement by NOAA, Reclamation, USACE, state and local agencies, and academics.

<sup>118</sup> Although the updated *Folsom Dam Water Control Manual* has some FIRO characteristics, USACE indicated that FIRO research was not explicitly used in the revision to the manual (personal communication from USACE staff to CRS staff, July 8, 2021).

<sup>119</sup> P.L. 116-260, §157. Also in Section 157 of P.L. 116-260, Congress directed that the report include an assessment of the viability of FIRO in two basins—Upper Missouri River and North Platte River. Section 8303 of WRDA 2022 (continued...)

published a report on the screening process to identify additional facilities suitable for FIRO and the results of the first of three rounds in the suitability screening process.<sup>120</sup> The first suitability screening eliminated 31% of USACE facilities. In the interim, Congress continued to support FIRO through direction related to FY2023 appropriations and additional authorizations for studying FIRO and performing pilot projects in Section 8304 of WRDA 2022. In Section 1162 of WRDA 2024, Congress authorized the Secretary of the Army to incorporate FIRO into the updates of water-control manuals for USACE-owned or -operated reservoirs.

## **Drought Flexibilities and the Endangered Species Act**

Some federal and nonfederal water infrastructure projects in the United States must ensure that their operations are consistent with the requirements of the Endangered Species Act (ESA; 16 U.S.C. §§1531-1544) due to their potential to harm threatened or endangered species listed under the ESA, or adversely modify designated critical habitat.<sup>121</sup> Operational plans and ESA regulations often result in water projects being required to maintain a certain level of water in an ecosystem or river for listed species or mitigate the effects of a project on listed species. Under drought conditions, these requirements can lead to disagreements on how much water to allocate for species and habitat, and human water uses among stakeholders vying for scarce water supplies.

Operators of water projects may be required to undergo consultation with FWS or NMFS (collectively referred to as the Services) under Section 7 of the ESA. Section 7 of the ESA requires that federal agencies ensure their discretionary actions, including the actions of nonfederal parties granted a federal approval, permit, or funding, are “not likely to jeopardize the continued existence” of any endangered or threatened species, or adversely modify or destroy critical habitat.<sup>122</sup> The consultation requirements apply only when an action involves “discretionary Federal involvement or control,” not when actions are mandated by statute.<sup>123</sup> The consultation concludes with one of the Services issuing a *biological opinion* (BiOp) that imposes terms and conditions on project operations to conserve listed species and critical habitat. Some BiOps for water project operations also include policy mechanisms to allow water users to maximize water supplies during drought while continuing to conserve listed species. These mechanisms often include the use of adaptive management (AM), real-time management of water supplies, artificial propagation of listed species (e.g., hatcheries), and habitat restoration.

Project operators use AM to fine-tune operations during drought conditions. AM is the process of incorporating new scientific and programmatic information into the implementation of a project or plan to adapt to changing conditions that may be difficult to predict. AM promotes adaptable decisionmaking that modifies existing activities and/or creates new activities if new

---

added a third basin—Apalachicola Chattahoochee Flint River Basin. Congress authorized the Secretary to implement FIRO at reservoirs in the three basins if the assessment finds FIRO viable for the reservoirs. Congress required that if these operations are found to be viable, the Secretary of the Army, subject to the availability of appropriations, shall implement FIRO for at least one reservoir in each basin.

<sup>120</sup> Elissa M. Yeates et al., *The Forecast-Informed Reservoir Operations (FIRO) Screening Process: Stage A Development and National Results*, U.S. Army Engineer Research and Development Center, ERDC/CHL TR-26-17, July 2026, <https://erdc-library.erdc.dren.mil/server/api/core/bitstreams/4648b9ee-96d1-4904-b824-d7a02999d132/content>.

<sup>121</sup> For more information on the Endangered Species Act (ESA; 16 U.S.C. §§1531-1544), see CRS Report R46677, *The Endangered Species Act: Overview and Implementation*, by Pervaze A. Sheikh and Erin H. Ward; and CRS Report R46867, *Endangered Species Act (ESA) Section 7 Consultation and Infrastructure Projects*, by Erin H. Ward and Pervaze A. Sheikh.

<sup>122</sup> 16 U.S.C. §1536(a).

<sup>123</sup> 50 C.F.R. §402.03.

circumstances arise (e.g., new scientific information) or projects are not meeting their goals. Under drought conditions, AM provisions in a project's BiOp might allow federal agencies to address the conservation needs of a listed species while still accomplishing project purposes. In some cases, the operations of water infrastructure have used AM to account for changes in environmental conditions that were not contemplated in a BiOp. In these cases, the agency may seek concurrence from either FWS or NMFS to temporarily modify the BiOp's implementation to account for such unforeseen circumstances.

Some BiOps also allow for "real-time" management (i.e., management based on changing conditions) of water projects under certain circumstances, including drought, to maximize water supplies for users or address immediate threats to species. This approach allows managers to alter project operations in real time to respond to changing conditions. For example, pursuant to the 2019 BiOp for the coordinated operations of the Central Valley Project (CVP) and the State Water Project in California, project pumping operations may adhere to certain preestablished levels, unless findings from real-time monitoring of threatened Delta smelt trigger changes that aim to conserve the species. Congress authorized this approach for CVP operations under Section 4001 of the WIIN Act. At the time, supporters generally endorsed the provisions as short-term measures to alleviate the effects of the ongoing drought in California on users.<sup>124</sup>

## **Support for Nonfederal Drought Planning and Projects**

In addition to the aforementioned federal activities and programs, multiple federal agencies have programmatic authorities to support nonfederally led projects related to drought. Some of these authorities are discussed below.

### **Bureau of Reclamation Authorities<sup>125</sup>**

#### ***WaterSMART***

Congress has enacted multiple *programmatic* (i.e., not project-based) authorities for Reclamation to support nonfederal efforts to conserve water. In contrast to Reclamation's traditional activities, there is typically no federal ownership role associated with projects supported by these authorities. As with Reclamation's other authorities, most of its programmatic authorities are limited to activities to the 17 reclamation states and territories, as defined in statute.<sup>126</sup> Reclamation combines funding for its programs promoting water conservation into a single program—the WaterSMART (Sustain and Manage American Resources for Tomorrow) program.<sup>127</sup> This includes funding for Reclamation's drought-related authorities, as well as other programs that address nonfederal water supplies.

The WaterSMART Drought Response Program (DRP) is Reclamation's only program specifically dedicated to addressing drought. The DRP assists water managers with developing and implementing comprehensive drought plans and related projects that build long-term drought resiliency. Specific sub-program areas of the DRP include Contingency Planning, Resiliency Projects, and Emergency Response Actions. In an effort to incentivize advanced planning and

---

<sup>124</sup> For additional information about these provisions, see CRS Report R44986, *Water Infrastructure Improvements for the Nation (WIIN) Act: Bureau of Reclamation and California Water Provisions*, by Charles V. Stern, Pervaze A. Sheikh, and Nicole T. Carter.

<sup>125</sup> For more information on these programs, contact Charles V. Stern, Specialist in Natural Resources Policy.

<sup>126</sup> 43 U.S.C. §391.

<sup>127</sup> More information on Reclamation WaterSMART programs is available at <https://www.usbr.gov/watersmart/>.

mitigation, Reclamation directs the majority of DRP funding to the first two sub-program areas. The bureau typically funds DRP planning and resiliency projects sequentially. First, it funds planning for communities to learn how droughts may affect them, as well as to scope potential projects to reduce the impacts of the next drought. Subsequently, it prioritizes grant funding for resiliency projects identified through the planning process.

Generally, Reclamation supports resiliency projects that attempt to either (1) increase the reliability of water supplies by providing additional alternatives during drought (e.g., constructing new infrastructure, such as intakes or groundwater banking facilities) or (2) improve water management by providing entities with tools and decision support (e.g., improved modeling, access to water markets). Both contingency planning and resiliency efforts include a 50/50 cost share with nonfederal sponsors.

Reclamation reserves a small amount of DRP funding for emergency response actions, as authorized in the Reclamation States Drought Relief Act of 1991 (P.L. 102-250). Eligible projects include temporary construction activities (e.g., temporary pipes and pumps) and other actions authorized under Title I of the act that can be completed within a year, such as water purchases or use of Reclamation facilities to convey and store water. Reclamation conducts emergency response actions through contracts rather than by providing financial assistance, and approval of these actions is subject to a number of other requirements, such as a state or tribal drought declaration.<sup>128</sup>

Other WaterSMART programs also function to lessen drought impacts by supporting alternative water supplies, increased efficiency, and/or water resources conservation. For example, the Title XVI Program provides cost-shared financial assistance for authorized nonfederal water reuse or recycling projects.<sup>129</sup> Similarly, Reclamation's Desalination Program provides federal financial support for selected nonfederal desalination projects that Congress has approved.<sup>130</sup> Although project selection processes for both programs prioritize drought resiliency (among other criteria), they do not prioritize funding for drought-stricken areas. Generally, projects in the Title XVI Program and the Desalination Program take years to construct.

Several WaterSMART programs promote water conservation and related efforts and therefore provide benefits that may increase drought resiliency. For instance, WaterSMART Grants provide cost-shared federal funding for projects in multiple categories, including water and energy efficiency grants, small-scale water efficiency grants, and support for projects that benefit ecological values or promote watershed health. Separately, Reclamation's Basin Study Program supports efforts to address imbalances between water supply and demand in western river basins through applied science tools, guidance, and information to support water management planning. Reclamation's Cooperative Watershed Management Program provides funding to watershed groups to encourage stakeholders to find local solutions to water management needs. Most of these programs require some form of cost sharing from nonfederal sponsors to leverage federal funding, and most include caps on the amount of federal assistance.<sup>131</sup>

---

<sup>128</sup> This requirement applies only to entities without an approved DCP. For more information on emergency drought assistance requirements, see Reclamation, "Request for Emergency Drought Assistance Checklist," WTR 10-01, Appendix C, <https://www.usbr.gov/recman/wtr/wtr10-01-AppC.pdf>.

<sup>129</sup> For more information about this program, see Reclamation, "Title XVI – Water Reclamation and Reuse," <https://www.usbr.gov/watersmart/title/index.html>.

<sup>130</sup> For more information about this program, see Reclamation, "Desalination," <https://www.usbr.gov/watersmart/desalination/index.html>.

<sup>131</sup> For more information on these and other Reclamation programs supporting nonfederal water supplies and planning, see Reclamation, "WaterSMART," <https://www.usbr.gov/watersmart/>.

## ***Geographically Targeted Drought Support***

Congress also has authorized targeted grant programs to combat drought. For example, in Section 206 of the Energy and Water Development and Related Agencies Appropriations Act, 2015 (P.L. 113-235), Congress authorized Reclamation to fund conservation projects in the Colorado River Basin that add overall water supply to the Colorado River Basin system.<sup>132</sup> The resulting program, known as the Pilot System Conservation Program, has funded voluntary conservation projects and reductions of water use in in Lake Mead and other Colorado River system reservoirs. Congress authorized and funded additional targeted efforts addressing Colorado River Basin drought in the Infrastructure Investment and Jobs Act (IIJA; P.L. 117-58). Congress has also authorized Reclamation to provide targeted drought support in specific river basins by paying irrigators to forgo deliveries of project water. Such efforts have been authorized and funded in both the Colorado River Basin in P.L. 117-169 (commonly referred to as the Inflation Reduction Act of 2022) and in the Klamath River Basin under P.L. 106-498, as amended.

## **USDA Programs and Authorities**

### ***Rural Utilities Service***<sup>133</sup>

USDA's Rural Utilities Service (RUS) provides grants, loans, loan guarantees, and technical assistance to support water systems in rural communities.<sup>134</sup> Some of these programs are tailored for emergency situations, whereas others prioritize awards for rural communities facing drought-related declines in water quantity or quality. For most RUS loans and grants, rural areas are areas with populations of 10,000 or fewer people.<sup>135</sup> For RUS loan guarantee programs, rural communities are areas with populations of 50,000 or fewer people.<sup>136</sup> The RUS programs that may assist in addressing drought-related rural water issues include the following:

- **Emergency Community Water Assistance Grant Program.**<sup>137</sup> This program provides grants to help eligible rural areas prepare for, or recover from, emergencies that threaten the availability of safe, reliable drinking water. Emergencies include natural disasters such as droughts. A federal disaster declaration is not required to participate in this program. Grants of up to \$150,000 are available for water transmission line projects that construct waterline extensions, repair breaks in existing water distribution lines, and other maintenance activities necessary to replenish the municipal water supply. Grants of up to \$1 million are available for water source projects such as constructing new wells, reservoirs, and treatments plants.

---

<sup>132</sup> For more information about ongoing drought in the Colorado River Basin and federal response efforts, see CRS Report R45546, *Management of the Colorado River: Water Allocations, Drought, and the Federal Role*, by Charles V. Stern and Pervaze A. Sheikh.

<sup>133</sup> For more information about the Rural Utilities Service, contact Zachary T. Neuhofer, Analyst in Agricultural Policy.

<sup>134</sup> RUS loans are issued from RUS to the borrowers. For RUS loan guarantees, RUS guarantees the loans issued by eligible lenders. The eligible lenders loans to qualified borrowers.

<sup>135</sup> 7 U.S.C. §1991(a)(13)(B).

<sup>136</sup> 7 U.S.C. §1991(a)(13)(A).

<sup>137</sup> For additional information, see USDA, Rural Development, "Emergency Community Water Assistance Grants," <https://www.rd.usda.gov/programs-services/emergency-community-water-assistance-grants>.

- **Rural Decentralized Water Systems Grant Program.**<sup>138</sup> This program provides assistance to help qualified rural homeowners to construct, refurbish, or repair individually owned wells and septic systems. The program provides grants to eligible nonprofit organizations to establish revolving loan funds. These loan funds issue loans or subgrants to rural homeowners for water well or septic systems projects. Loans of up to \$15,000 are available to homeowners. Some of this program's funds may be used to assist drought-affected households, although responding to drought is not an explicit program purpose.
- **Water and Waste Disposal Loan and Grant Program.**<sup>139</sup> This program provides loans and grants to eligible entities to construct new and improve existing water and waste disposal systems in eligible rural areas. Water systems include drinking water sourcing, treatment, storage, and distribution. It also includes storm water collection, transmission, and disposal. Systems affected by natural disasters, such as drought, may receive a priority.<sup>140</sup>
- **Water and Waste Disposal Loan Guarantee Program.**<sup>141</sup> This program provides loan guarantees to eligible entities to construct new and improve existing water and waste disposal systems in eligible rural areas. Systems affected by natural disasters, such as drought, may receive a priority.<sup>142</sup>

### *Natural Resources Conservation Service*<sup>143</sup>

USDA's Natural Resources Conservation Service (NRCS) provides assistance for watershed activities under four closely related authorities. Most of these programs pay a percentage of the cost to install infrastructure or correct impairments, and they require a local project sponsor. The programs are permanently authorized but subject to appropriations. Applications may be filed at any local or state NRCS office.<sup>144</sup>

- **Watershed and Flood Prevention Operations (WFPO) Program.** The WFPO program consists of two authorities—the Watershed Protection and Flood Prevention Act of 1954 (P.L. 83-566) and the Flood Control Act of 1944 (P.L. 78-534). These acts authorize NRCS to provide technical and financial assistance to state and local organizations to plan and install measures to prevent erosion,

---

<sup>138</sup> For additional information, see the USDA program page at <https://www.rd.usda.gov/programs-services/rural-decentralized-water-systems-grant>.

<sup>139</sup> For additional information, see the USDA program page at <https://www.rd.usda.gov/programs-services/water-waste-disposal-loan-grant-program>.

<sup>140</sup> As per 7 C.F.R. §1780.17(g), the Rural Utilities Service Administrator may assign up to 15 additional priority points to applications to account for items such as geographic distribution of funds, the highest priority projects within a state, and emergency conditions caused by economic problems or natural disasters.

<sup>141</sup> For additional information, see the USDA program page at <https://www.rd.usda.gov/programs-services/water-environmental-programs/water-waste-disposal-loan-guarantees>.

<sup>142</sup> As per 7 C.F.R. §5001.317(f), the Rural Utilities Service Administrator may assign up to 15 additional priority points to applications to account for items such as geographic distribution of funds, the highest priority projects within a state, and emergency conditions caused by economic problems or natural disasters.

<sup>143</sup> For more information on NRCS programs, contact Megan Stubbs, Specialist in Agricultural Conservation and Natural Resources Policy.

<sup>144</sup> To find a local NRCS office, see <https://www.farmers.gov/working-with-us/service-center-locator>.

sedimentation, and flood damage. Projects may also include drought control and water supply measures to conserve, develop, and use land and water resources.<sup>145</sup>

- **Watershed Rehabilitation Program.** The Watershed Rehabilitation Program provides technical and financial assistance for planning, design, and implementation of activities to rehabilitate aging watershed dam projects (including upgrading or removing dams) in communities to address health and safety concerns. Eligible activities may include construction or augmentation of existing water supplies based on current and future water supply demands. Only dams constructed under the WFPO program are eligible.<sup>146</sup>
- **Emergency Watershed Protection (EWP) Program.** The EWP program assists local project sponsors in implementing emergency recovery measures for runoff retardation and erosion prevention to relieve imminent hazards to life and property created by natural disasters, including drought. For example, the program can be used to reseed drought-stricken areas that would be prone to erosion and could pose a threat to life or property.<sup>147</sup>

### **USACE Authorities<sup>148</sup>**

In 1974, Congress provided USACE with authority (33 U.S.C. §701n) to assist with emergency water supplies (e.g., bulk or bottled water) and their transport during drought when state resources are exceeded and a public health threat is imminent.<sup>149</sup> USACE provides assistance only to meet any minimum public health and welfare requirements that cannot be met in the immediate future by state or local actions or through reasonable conservation measures. USACE assistance may include a variety of activities. For some, the costs of the activity are initially borne by USACE and must be reimbursed to USACE by the recipient (i.e., the costs are ultimately 100% nonfederal). The costs of other activities are fully borne by USACE, (i.e., 100% federal), as follows:

- Purchasing or acquiring the water, acquiring and operating the storage facility at the terminal point, and constructing permanent water facilities (100% nonfederal);
- USACE well construction costs (100% nonfederal); and
- Water transport costs (100% federal).

Funding for actions pursuant to this authority is provided through the USACE Flood Control and Coastal Emergencies Account.

---

<sup>145</sup> For additional information on the Watershed and Flood Prevention Operations Program, see CRS Report R46471, *Federally Supported Projects and Programs for Wastewater, Drinking Water, and Water Supply Infrastructure*, coordinated by Jonathan L. Ramseur.

<sup>146</sup> The Watershed Rehabilitation Program is authorized under Section 313 of the Grain Standards and Warehouse Improvement Act of 2000 (P.L. 106-472) and Section 14 of the Watershed Protection and Flood Prevention Act, as amended (16 U.S.C. §1012). For more information, see CRS Report R47383, *Federal Assistance for Nonfederal Dam Safety*, coordinated by Anna E. Normand.

<sup>147</sup> The Emergency Watershed Protection Program is authorized under Title IV of the Agricultural Credit Act of 1978 (P.L. 95-334; 16 U.S.C. §2203) and 33 U.S.C. §701b-1. For additional information on the program, see CRS Report R42854, *Emergency Assistance for Agricultural Land Rehabilitation*, by Megan Stubbs.

<sup>148</sup> For more information on these authorities, contact Nicole T. Carter, Specialist in Natural Resources Policy.

<sup>149</sup> This authority cannot be used for the provision of water for livestock, irrigation, recreation, or commercial/industrial use. Eligible entities are limited to drought-distressed political subdivisions, farmers, and ranchers.

A governor, a governor's representative, or the governing body of a Tribe must make a written Central Valley Project request for assistance to USACE. The USACE Director of Civil Works or the Assistant Secretary of the Army (Civil Works) makes the determination whether an area has an inadequate water supply that is causing, or is likely to cause, a substantial threat to the health and welfare of the area's inhabitants.

Separate from this emergency water supply authority, USACE has the authority to contract for limited quantities of water (if available) from its reservoirs for municipal and industrial purposes. Congress also has provided USACE with a technical assistance authority for state and local entities, referred to as Planning Assistance to States (PAS). Under the PAS authority, USACE has worked with state and local governments in Texas, Virginia, and Iowa to support planning for drought conditions.

### **U.S. Environmental Protection Agency Programs<sup>150</sup>**

Improving public water systems' resilience to droughts and to other events that may disrupt the provision of safe and reliable water supplies has been a focus of congressional attention regarding programs authorized under the Safe Drinking Water Act (SDWA), and other EPA authorities. Congress has established programs, which are administered by EPA, that provide financial assistance to public water systems for projects that help ensure the provision of a safe and reliable water supply. Further, several provisions of the Safe Drinking Water Act promote water systems' preparedness for, and resilience to, events that may disrupt water service (e.g., natural hazards, malevolent acts).<sup>151</sup>

### ***Drinking Water State Revolving Fund<sup>152</sup>***

In 1996, Congress established the Drinking Water State Revolving Fund (DWSRF) program to provide financial assistance to public water systems for infrastructure projects needed to comply with federal drinking water regulations and protect public health.<sup>153</sup> Projects that aim to increase water system resilience to drought through the development of a new or alternative drinking water source(s), and/or through the construction or rehabilitation of water storage, are eligible for DWSRF financial assistance.<sup>154</sup>

Congress annually appropriates DWSRF program funding, which EPA allots among the states as grants to capitalize their drinking water state revolving loan funds. Every year, each state must match 20% of its annual capitalization grant and develop an intended-use plan for the allotted funds. The SDWA requires states to prioritize projects that address the most serious human health risks, are necessary to ensure regulatory compliance, and assist systems most in need on a per household basis, according to state affordability criteria. Depending on the applicants' project types and the state's individual circumstances, a state may choose to prioritize DWSRF assistance for drought resilience projects, such as alternative water projects to replace diminished or

---

<sup>150</sup> For more information on these programs, contact Elena H. Humphreys, Specialist in Environmental Policy.

<sup>151</sup> 42 U.S.C. §§300f et seq. For a discussion of Safe Drinking Water Act (SDWA; 42 U.S.C. §300j-12) risk and resilience assessments and emergency response planning requirements, see CRS In Focus IF11777, *Safe Drinking Water Act (SDWA): Water System Security and Resilience Provisions*, by Elena H. Humphreys.

<sup>152</sup> For more information about the Drinking Water State Revolving Fund (DWSRF), see CRS Report R47935, *Changes to the Drinking Water State Revolving Fund (DWSRF) Program*, by Elena H. Humphreys.

<sup>153</sup> The Safe Drinking Water Act Amendments of 1996 (P.L. 104-182), Section 130, added the DWSRF provisions (§1452) to the SDWA. The key purpose of DWSRF financial assistance is to support projects that the Environmental Protection Agency (EPA) has determined through guidance will facilitate compliance with SDWA.

<sup>154</sup> EPA, *Drinking Water State Revolving Fund Eligibility Handbook*, EPA 816-B-17-001, June 2017.

contaminated water sources. While the primary type of DWSRF financial assistance is low-interest-rate loans, SDWA Section 1452 authorizes states to provide additional subsidization (including forgiveness of principal) to disadvantaged communities.<sup>155</sup> The federal capitalization grants, together with state funds (e.g., state match, loan repayments, leveraged bonds, and other state sources), are intended to build a sustainable source of drinking water infrastructure funding.

### ***Drinking Water System Infrastructure Resilience and Sustainability Program***

In 2018, America's Water Infrastructure Act of 2018 authorized the Drinking Water System Infrastructure Resilience and Sustainability Program.<sup>156</sup> This program authorized EPA to establish a grant program for small and disadvantaged public water systems. This program directs EPA to award funds to eligible public water systems for projects that increase resilience to natural hazards, including hydrologic changes.<sup>157</sup> Eligible projects include those that increase water use efficiency, enhance water supply through watershed management or desalination, and increase energy efficiency in the conveyance or treatment of drinking water. Congress first provided appropriations for this grant program in FY2020.<sup>158</sup> In 2021, the IIJA reauthorized appropriations of \$25.0 million for each year from FY2022 through FY2026 for this program and added a 10% cost share.<sup>159</sup>

### ***Water Infrastructure Finance and Innovation Act<sup>160</sup>***

The Water Resources Reform and Development Act of 2014 (P.L. 113-121) authorized the Water Infrastructure Finance and Innovation Act (WIFIA) program to promote development of and private investment in water infrastructure projects.<sup>161</sup> WIFIA authorized EPA to provide credit assistance in the form of secured or direct loans for a range of water infrastructure projects.<sup>162</sup>

---

<sup>155</sup> SDWA §1452(d); 42 U.S.C. §300j-12(d) authorized states to provide additional subsidization to disadvantaged communities. *Disadvantaged community* is defined as the service area of a public water system that meets affordability criteria developed by the state.

<sup>156</sup> America's Water Infrastructure Act of 2018 (P.L. 115-270) Section 2005 amended SDWA §1459A to add this grant program.

<sup>157</sup> Eligible entities are defined in SDWA §1459A(c)(2)(B). *Natural hazard* is defined as a natural event that threatens the functioning of a community water system, including an earthquake, tornado, flood, hurricane, wildfire, and hydrologic changes (SDWA §1433(h)(2)).

<sup>158</sup> The Further Consolidated Appropriations Act, 2020 (P.L. 116-94) provided \$3.0 million for this grant program.

<sup>159</sup> IIJA §50104.

<sup>160</sup> For more information about the Water Infrastructure Finance and Innovation Act (WIFIA) program, see CRS Report R46471, *Federally Supported Projects and Programs for Wastewater, Drinking Water, and Water Supply Infrastructure*, coordinated by Jonathan L. Ramseur.

<sup>161</sup> 33 U.S.C. §§3901-3914.

<sup>162</sup> WIFIA also authorized a loan program for USACE. As of this report, USACE's credit assistance program (referred to as the Corps Water Infrastructure Financing Program, CWIFP) had not made loans; USACE solicited preliminary nonfederal CWIFP applications for nonfederal dam safety projects in 2023, and in 2024 invited 18 prospective borrowers to complete their applications for an estimated almost \$3.19 billion in loans. For more information on the USACE program, see CRS Insight IN12021, *Corps Water Infrastructure Financing Program (CWIFP)*, by Nicole T. Carter.

WIFIA-eligible drinking water projects include those projects eligible for DWSRF financial assistance as well as other projects that may support drought resilience, such as through the following activities:

- Desalination;
- Aquifer recharge or development of alternative water supplies to reduce aquifer depletion;
- Water recycling and/or reuse; and
- Mitigation, prevention, or reduction of the effects of drought.

Entities eligible for WIFIA assistance include (1) state infrastructure financing authorities; (2) corporations; (3) partnerships; (4) joint ventures; (5) trusts; and (6) federal, state, local, or tribal governments or instrumentalities. WIFIA establishes broad selection criteria that EPA uses to rank projects, including a project's national or regional significance with respect to economic and public benefits, creditworthiness, and readiness.

Each year that Congress appropriates funds to cover subsidies for WIFIA loans, EPA publishes a Notice of Funding Availability (NOFA) to provide interested entities with WIFIA application information. In each NOFA, EPA identifies considerations for project prioritization (e.g., repairing aging infrastructure). EPA issued a NOFA in November 2025 listing priorities for WIFIA. One of EPA's funding priorities includes "projects that incorporate new or innovative technology," such as water reuse, which can alleviate the "effects of drought."<sup>163</sup> This NOFA also included the criteria for project selection, such as "the extent to which a project serves regions with significant water resource challenges."<sup>164</sup>

### ***WaterSense***

WaterSense is a voluntary labeling program that EPA created in 2006 to encourage the development and use of water-efficient products and services. Programs like WaterSense identify and promote the use of water-efficient products and/or services that may lessen the effects of drought through means other than providing financial assistance. Through WaterSense, EPA develops water-efficiency specifications for products, certain services, and homes; licenses third-party certification bodies; and maintains a registry of WaterSense-labeled products and certified services. The 115<sup>th</sup> Congress authorized and expanded WaterSense in Section 4306 of America's Water Infrastructure Act of 2018 (P.L. 115-270).<sup>165</sup>

EPA has issued WaterSense specifications for categories of services and a variety of products, including residential toilets, showerheads, bathroom faucets, commercial toilets, urinals, irrigation controllers, and spray sprinkler bodies. To obtain certification to use a WaterSense label, manufacturers must develop products that meet EPA specifications. EPA states that a water-efficient product should generally (1) reduce water use by at least 20% from federally mandated water-use conservation standards and (2) function at least as well as regular models. For products without federal standards, such as irrigation equipment, WaterSense certifications are based on calculations of average efficiency. The use of water-efficient products may reduce demand for

---

<sup>163</sup> EPA, "Notice of Funding Availability for Credit Assistance Under the Water Infrastructure Finance and Innovation Act (WIFIA) Program," 90 *Federal Register* 54671-54675, November 28, 2025.

<sup>164</sup> 90 *Federal Register* 54675, November 28, 2025.

<sup>165</sup> 42 U.S.C. §6294b.

water, helping communities improve drought resilience. In 2024, EPA estimated that the program saved 9.9 trillion gallons of water from 2006 to 2024.<sup>166</sup>

### **Selected Federal Drought Programs for Tribes<sup>167</sup>**

Many *federally recognized Tribes* and Alaska Native villages (ANVs) experience the effects of drought on their lands.<sup>168</sup> Federal agencies have a unique relationship with Tribes due to the *federal trust responsibility*, which is a legal obligation under which the United States, through acts of Congress and court decisions, “has charged itself with moral obligations of the highest responsibility and trust” toward Tribes.<sup>169</sup> Federal agencies work with Tribes in different ways (e.g., tribal consultation, grants or cooperative agreements, technical assistance, monitoring, research). Tribes are eligible for many federal drought assistance programs available to state and local governments, as outlined elsewhere in this report. Tribes are also eligible for specific programs and funding that can be used to prepare for or address drought impacts on tribal lands and waters, some of which are described below.<sup>170</sup>

### ***U.S. Department of Agriculture***

Many Tribes are located in rural areas and, as such, are eligible for certain USDA Rural Development infrastructure programs. In addition, USDA has selected tribal-specific programs:

- **USDA Grants for Rural and Native Alaskan Villages.** These grants are to help ANVs provide safe drinking water and waste disposal systems to households and businesses.<sup>171</sup> ANVs can use the grants for projects addressing droughts and other natural disasters, when those disasters cause dire sanitation conditions.<sup>172</sup>
- **Water and Waste Facility Loans and Grants to Alleviate Health Risks for Colonias.** These grants are to help rural communities with populations of 10,000 or fewer people in areas recognized as colonias facing significant health risks resulting from lack of access to safe, reliable drinking water and waste disposal facilities. *Colonias* are state-designated, low-income, unincorporated areas along

---

<sup>166</sup> EPA, *WaterSense 2024 Accomplishments Report*, EPA-832-F-25-008.

<sup>167</sup> For more information on tribal natural resource issues, contact Mariel J. Murray, Specialist in Natural Resources Policy.

<sup>168</sup> A *federally recognized Tribe* (Tribe) is an entity formally recognized as having a government-to-government relationship with the United States, entailing special rights, immunities, and privileges as well as eligibility for certain federal programs and services (25 C.F.R. §83). Alaska Native village (ANV) is defined in the Alaska Native Claims Settlement Act (43 U.S.C. §1602) as “any tribe, band, clan, group, village, community, or association in Alaska listed in sections 1610 and 1615 of this title, or which meets the requirements of this chapter, and which the Secretary determines was, on the 1970 census enumeration date (as shown by the census or other evidence satisfactory to the Secretary, who shall make findings of fact in each instance), composed of twenty-five or more Natives.”

<sup>169</sup> *Seminole Nation v. U.S.*, 316 U.S. 286, 296-297 (1942). For a general overview of the trust relationship, see *U.S. v. Jicarilla Apache Nation*, 564 U.S. 162 (2011).

<sup>170</sup> This is not a comprehensive list of programs available to Tribes.

<sup>171</sup> USDA Rural Development, “Grants for Rural and Native Alaskan Villages,” <https://www.rd.usda.gov/programs-services/water-environmental-programs/grants-rural-and-native-alaskan-villages>.

<sup>172</sup> As per 7 C.F.R. §1784.2, dire sanitation conditions mean (1) recurring instances of illness reasonably attributed to waterborne communicable disease have been documented or insufficient access to clean water creates a persistent threat of water-washed diseases; (2) no community-wide water and sewer system exists and individual residents must haul water to or human waste from their homes and/or use pit privies; or (3) an appropriate federal agency (such as the Centers for Disease Control and Prevention) or regulatory Agency of the State of Alaska determines that the drinking water and/or sewer system does not meet current regulatory requirements.

the U.S.-Mexico border that lack basic infrastructure, especially for drinking water and wastewater.<sup>173</sup> Funding may be prioritized for projects addressing droughts and other natural disasters.<sup>174</sup>

- **Water and Waste Facility Loans and Grants to Alleviate Health Risks on Tribal Lands.** These grants are to help rural communities with populations of 10,000 or fewer people on tribal lands facing significant health risks resulting from lack of access to safe, reliable drinking water and waste disposal facilities.<sup>175</sup> Funding may be prioritized for projects addressing droughts and other natural disasters.<sup>176</sup>

## **USACE**

Since 2000, under USACE's Tribal Partnership Program authority, USACE has been authorized to assist Tribes to become more drought resilient.<sup>177</sup> Congress has expanded the Tribal Partnership Program's authority from a program for technical assistance to a program that also may support project design and construction. For example, Congress in 2018 and in subsequent amendments has authorized USACE to undertake construction of certain water projects with Tribes.<sup>178</sup> These plans and projects can range from environmental restoration actions to build resilience and protect important cultural landscapes to infrastructure investments to improve water supply reliability.<sup>179</sup>

## **Department of the Interior**

Various agencies in the Department of the Interior are authorized to assist Tribes. Below is a selected set of the DOI agencies and their authorities and activities relevant to tribal efforts to address drought impacts on tribal lands and waters, among other eligible activities.

The Snyder Act of 1921 authorizes the Bureau of Indian Affairs (BIA) to operate programs, and spend federal funds, for the benefit and assistance of Tribes and individual tribal members. Under

---

<sup>173</sup> 7 C.F.R. §1777.4. For more information, see that USDA program page at <https://www.rd.usda.gov/programs-services/water-environmental-programs/water-and-waste-facility-loans-and-grants-alleviate-health-risks-colonias>.

<sup>174</sup> As per 7 C.F.R. §1777.33(c)(9)(i), the state Rural Development director or his or her designee, with written justification, may prioritize funding for selected items including natural disasters.

<sup>175</sup> For more information, see the USDA program page at <https://www.rd.usda.gov/programs-services/water-environmental-programs/water-and-waste-facility-loans-and-grants-alleviate-health-risks-tribal-lands>.

<sup>176</sup> As per 7 C.F.R. §1777.33(c)(9)(i), the state Rural Development director or his or her designee, with written justification, may prioritize funding for selected items including natural disasters.

<sup>177</sup> 33 U.S.C. §2269.

<sup>178</sup> P.L. 115-270, Title I, §1157(i); P.L. 116-260, Division. AA, Title III, §303; P.L. 117-263, Division H, Title LXXXI, §8111.

<sup>179</sup> The Tribal Partnership Program's authority provides for no nonfederal cost sharing for planning and technical assistance activities. The program's authority provides for nonfederal cost sharing for construction projects (which varies depending on the project purpose) and notes that cost sharing agreements under the program are subject to a Tribe's ability to pay, as determined by the Secretary of the Army (33 U.S.C. §2269(d)(1)(A) and 33 U.S.C. §2269(d)(4)(B)). The nonfederal cost share under the program can be met through in-kind services (33 U.S.C. §2269(d)(2)). USACE construction projects performed pursuant to the program's authority do not require congressional project authorization if their federal costs are less than \$26 million. Separately, Congress waived a portion of most USACE study and project costs for Tribes (33 U.S.C. §2310; the amount of the waiver is adjusted annually with inflation).

that general authority and others, BIA supports a wide variety of activities, such as water management planning and drought resilience, as follows:<sup>180</sup>

- **Water Management, Planning and Pre-Development.** BIA performs technical studies to help Tribes gather hydrology data, develop best practices to use water, create drought management plans that include water conservation techniques, and undertake projects to support Indian water rights.<sup>181</sup>
- **Construction.** Construction activities occur under multiple authorities and include construction of Indian irrigation and power systems, as well as construction associated with the safety of dams on Indian reservations, BIA-owned drinking water and sanitation systems, and other facilities.<sup>182</sup>
- **Tribal Community Resilience Awards.** These awards support Tribes in building resilience to various threats. Potential activities include “community adaptation planning, community-led relocation, managed retreat (i.e., partial relocation), protect-in-place efforts, and ocean and coastal management.”<sup>183</sup> In 2024, several Tribes received funding to address drought risk.<sup>184</sup>

The Bureau of Reclamation has several programs specific to Tribes in addition to its more general authorities addressing drought. For example, the Native American Affairs Program provides technical support for Indian water rights settlements, and assists tribal governments with developing, managing, and protecting their water resources.<sup>185</sup> Separately, individual Indian water rights settlements between the federal government, Tribes, and others provide federal funding and support to develop tribal water rights, often in water-stressed areas.<sup>186</sup>

---

<sup>180</sup> Snyder Act, P.L. 67-85, 42 Stat. 208 (1921). The Bureau of Indian Affairs (BIA)’s mission is “to enhance the quality of life, promote economic opportunity, and carry out the responsibility to protect and improve the trust assets of American Indians and Alaska Natives.” (U.S. Department of the Interior- Indian Affairs, “Bureau of Indian Affairs (BIA),” <https://www.bia.gov/bia>).

<sup>181</sup> U.S. Department of the Interior, The United States Department of the Interior Budget Justifications and Performance Information for Fiscal Year 2025- Bureau of Indian Affairs, p. IA-TNR-34, [https://www.bia.gov/sites/default/files/media\\_document/fy2025-508-bia-greenbook.pdf](https://www.bia.gov/sites/default/files/media_document/fy2025-508-bia-greenbook.pdf).

<sup>182</sup> U.S. Department of the Interior, The United States Department of the Interior Budget Justifications and Performance Information for Fiscal Year 2025- Bureau of Indian Affairs, pp. IA-ES-6 and IA-CON-RM-4.

<sup>183</sup> BIA, “Tribal Community Resilience Annual Awards Dashboard,” <https://experience.arcgis.com/experience/49f7458ad70c4bac949f421ddf5081c3/>.

<sup>184</sup> BIA, “BIA-Tribal Community Resilience, FY24 Annual Awards Program Summary,” [https://www.bia.gov/sites/default/files/media\\_document/fy24\\_tcr\\_aap\\_awards\\_summary.pdf](https://www.bia.gov/sites/default/files/media_document/fy24_tcr_aap_awards_summary.pdf)

<sup>185</sup> Bureau of Reclamation, “Native American Affairs Program,” <https://www.usbr.gov/native/programs/TAPprogram.html>. Reclamation cites numerous multiple authorities for this program.

<sup>186</sup> For more information see CRS Report R44148, *Indian Water Rights Settlements*, by Charles V. Stern.

## U.S. Environmental Protection Agency<sup>187</sup>

- **DWSRF Tribal Set-Aside.** Of the amounts appropriated annually for DWSRF, Congress authorizes EPA to set aside as much as 2% for grants for Indian tribes and ANVs.<sup>188</sup> Colonias, Indian tribes (including state-recognized tribes), and ANVs can apply to state DWSRFs.<sup>189</sup> EPA's list of the types of projects eligible for funding under the federal Drinking Water Tribal Set-Aside program includes projects to rehabilitate or develop sources of drinking water.<sup>190</sup> For tribal applicants to state DWSRFs, a state may prioritize DWSRF assistance for drought resilience projects, such as alternative water projects to replace diminished or contaminated water sources.
- **Indian Reservation Drinking Water Grant Program.** America's Water Infrastructure Act of 2018 (P.L. 115-270) authorized a grant program at EPA for public water systems that serve Indian tribes, called the Indian Reservation Drinking Water Program. IIRWA revised the program directing EPA to establish this grant program and expanded the eligible uses of the funds to either drinking water infrastructure improvements or wastewater system improvements. The Consolidated Appropriations Act, 2023 (P.L. 117-328) provided the first appropriation of \$4 million for FY2023 for this program. For FY2026, the Commerce, Justice, Science; Energy and Water Development; and Interior and Environment Appropriations Act, 2026 (P.L. 119-74) provided \$4 million for this grant program. The extent to which this funding will be used for water infrastructure projects to address drought is unclear.

## Recent Supplemental Funding Responding to Drought

Congress regularly appropriates discretionary annual funding for many of the drought programs, as discussed in the program-specific discussions above. Congress has also appropriated additional and supplemental funding for these programs in response to specific drought events, as well as new drought-related priorities. In some cases, these additional funds have significantly augmented certain drought-related activities. Several of these bills, enacted since 2020, are discussed below.

<sup>187</sup> For more information on these programs, contact Elena H. Humphreys, Analyst in Environmental Policy.

<sup>188</sup> The term *Indian tribe* is defined to include any Indian tribe having a federally recognized governing body carrying out substantial governmental duties and powers over any area and ANVs (42 U.S.C. §300f(14)). Under Section 1452(i) (42 U.S.C. §300j-12(i)) of the Safe Drinking Water Act, EPA may use 1.5% of the amounts appropriated annually to make grants to Indian tribes and ANVs. Starting in FY2010 (the Department of the Interior, Environment, and Related Agencies Appropriations Act, 2010, P.L. 111-88), Congress has authorized EPA to reserve up to 2.0% of the appropriated funds for Indian tribes and ANVs and has continued through the terms and conditions of subsequent appropriations. For funding information, see EPA, "Tribal Drinking Water Program Fund Allotments," <https://www.epa.gov/tribaldrinkingwater/tribal-drinking-water-program-fund-allotments>.

<sup>189</sup> The Administration for Native Americans defines *state recognized tribes* as Indian tribes and heritage groups that are recognized by individual states for their various internal state government purposes.

<sup>190</sup> EPA, "Tribal Drinking Water Program Eligibility Requirements," <https://www.epa.gov/tribaldrinkingwater/drinking-water-infrastructure-grants-tribal-set-aside-program>.

## Infrastructure Investment and Jobs Act

In 2021 under the IIJA (P.L. 117-58), Congress approved funding for several nationwide and regional drought-related provisions. Selected IIJA appropriations for drought-related activities, included funding for

- NRCS (\$918 million for NRCS watershed programs (e.g., the WFPO, Watershed Rehabilitation, and EWP Programs));<sup>191</sup>
- NOAA (\$492 million to NOAA for coastal and inland flood and inundation mapping and forecasting and for next-generation water modeling activities,<sup>192</sup> \$25 million for data acquisition and \$1 million for the study of a soil moisture and snowpack monitoring pilot program in the Upper Missouri River Basin,<sup>193</sup> and \$80 million for research supercomputing to improve climate and weather modeling capabilities related to drought, flood and wildfire prediction, detection, and forecasting);<sup>194</sup>
- USACE (\$40 million for Upper Missouri River Basin soil moisture and snowpack monitoring);<sup>195</sup>
- Reclamation (\$1.15 billion for new and augmented surface and groundwater storage projects in the western United States, \$1 billion for water reuse and recycling, \$250 million for eligible desalination projects, \$400 million for water and energy efficiency grants, and \$300 million for Reclamation Colorado River Basin funding<sup>196</sup>);<sup>197</sup> and
- EPA (\$11.7 billion for the DWSRF that could support public water system resilience to drought, among a range of eligible projects).<sup>198</sup>

## Extending Government Funding and Delivering Emergency Assistance Act

For six of the fiscal years between FY2018-FY2025, Congress appropriated funds for agricultural losses due to natural disaster through an annual supplemental appropriation. From FY2020-

---

<sup>191</sup> Division J, Title I. For additional information, see CRS In Focus IF11990, *Infrastructure Investment and Jobs Act (IIJA): Funding for USDA Broadband, Watershed, and Bioproduct Programs*, by Lisa S. Benson, Megan Stubbs, and Kelsi Braemort.

<sup>192</sup> For more information about these activities, see NOAA, “Research supercomputing,” <https://www.noaa.gov/infrastructure-law/infrastructure-law-climate-data-and-services/research-supercomputing>, and NOAA, “Flood and inundation mapping and forecasting,” <https://www.noaa.gov/infrastructure-law/infrastructure-law-climate-data-and-services/flood-and-inundation-mapping-and-forecasting>.

<sup>193</sup> For more information about these activities, see NOAA, “Water Resources Development Act data acquisition” <https://www.noaa.gov/infrastructure-law/infrastructure-law-climate-data-and-services/water-resources-development-act-data-acquisition>.

<sup>194</sup> Division J, Title II.

<sup>195</sup> Division J, Title III. For a December 2025 description of USACE work, see USACE, *Upper Missouri River Basin: UMRB – Plains Snow and Social Moisture Monitoring Network*, December 17, 2025, [https://www.nwd-mr.usace.army.mil/rcc/reports/pdfs/umb\\_mon\\_network\\_factsheet.pdf](https://www.nwd-mr.usace.army.mil/rcc/reports/pdfs/umb_mon_network_factsheet.pdf).

<sup>196</sup> For more information about Colorado River Drought authorities and funding, see CRS Report R45546, *Management of the Colorado River: Water Allocations, Drought, and the Federal Role*, by Charles V. Stern and Pervaze A. Sheikh.

<sup>197</sup> Division J, Title III.

<sup>198</sup> Division J, Title VI. EPA, *Drinking Water State Revolving Fund Eligibility Handbook*, EPA 816-B-17-001, June 2017.

FY2025, Congress included drought as a qualifying natural disaster in the supplemental appropriation laws.<sup>199</sup> Congress expanded the determination of a qualifying drought to include more types of drought conditions in the FY2022 Division B of the Extending Government Funding and Delivering Emergency Assistance Act (P.L. 117-43).<sup>200</sup> In addition, this law included \$750 million to compensate livestock owners for losses in calendar year 2021 due to drought and wildfires.<sup>201</sup> P.L. 117-43 also included drought-related funding for Reclamation. The act provided Reclamation with \$200 million in addition to regular appropriations for drought-related programs and projects. Reclamation allocated these funds to various projects and programs, including funding for drought resiliency activities at the California Central Valley Project (\$62 million) and the Colorado River Basin (\$45 million), as well as \$20 million for the Drought Response Program, among other activities.<sup>202</sup>

## **Inflation Reduction Act of 2022**

In P.L. 117-169, a budget reconciliation measure commonly referred to as the Inflation Reduction Act of 2022 (IRA 2022), Congress provided additional appropriations for drought-related activities to several agencies. For instance, in Section 50233 of the bill, Congress provided Reclamation with \$4.0 billion in FY2022 (expiring at the end of FY2026) to mitigate the impact of drought in reclamation states, with priority given to projects in the Colorado River basin. Additional provisions provided funding to multiple agencies for activities indirectly related to drought, such as research and development infrastructure and scientific equipment (e.g., Sections 40004 and 40005).

IRA 2022 also provided nearly \$20 billion in additional funding to the USDA agricultural conservation programs.<sup>203</sup> These conservation programs can address drought-related concerns as part of their broad purpose (see “Conservation”). Additional funds provided through IRA 2022 were directed to climate change-related goals, specifically greenhouse gas mitigation activities. The FY2025 budget reconciliation law (P.L. 119-21) rescinded and repurposed unobligated IRA 2022 funding for USDA agricultural conservation programs.<sup>204</sup> The repurposed funding was distributed to the same agricultural conservation programs named in the 2022 IRA, as well as

---

<sup>199</sup> Congress did not include drought as an eligible natural disaster for ad hoc agricultural assistance in the Bipartisan Budget Act of 2018 (P.L. 115-123, Division B, Title I) and in the Additional Supplemental Appropriations for Disaster Relief Act of 2019 (P.L. 116-20, Title I).

<sup>200</sup> In Section 791 of Division B of the Further Consolidated Appropriations Act, 2020 (P.L. 116-94), Congress required that eligible drought include counties with a U.S. Drought Monitor classification of D3 (extreme) or D4 (exceptional) in calendar years 2018 and 2019. In Division B of the Extending Government Funding and Delivering Emergency Assistance Act (P.L. 117-43), qualifying losses related to drought were expanded to include counties with D2 (severe) classification for eight consecutive weeks or D3 or higher classification according to the U.S. Drought Monitor.

<sup>201</sup> USDA implemented the 2022 supplemental appropriation through the creation of the Emergency Relief Program and the Emergency Livestock Relief Program. For more information, see USDA, FSA, “Emergency Relief,” <https://www.fsa.usda.gov/programs-and-services/emergency-relief/index>.

<sup>202</sup> For a complete list of allocations, see Bureau of Reclamation, FY2022 - Distribution of Additional Funding in P.L. 117-43, <https://www.usbr.gov/budget/2022/FY-2022-Extending-Government-Funding-and-Delivering-Emergency-Assistance-Act-Funding-Allocation-Distribution-List.pdf>.

<sup>203</sup> Funding was specifically provided in P.L. 117-169 for the Environmental Quality Incentives Program (EQIP), Conservation Stewardship Program (CSP), Agricultural Conservation Easement Program (ACEP), and Regional Conservation Partnership Program (RCPP).

<sup>204</sup> For additional information, see CRS In Focus IF13114, *Agricultural Conservation After Enactment of the FY2025 Budget Reconciliation Law (P.L. 119-21)*, by Megan Stubbs.

other programs such as USDA watershed programs.<sup>205</sup> The repurposed funding removed the climate change-specific direction.

## **Disaster Relief Supplemental Appropriations Act, 2023**

The Disaster Relief Supplemental Appropriations Act, 2023 (passed as Division N of the Consolidated Appropriations Act, 2023 [P.L. 117-328]) included emergency appropriations related to drought. Specifically, the 2023 supplemental appropriations provided \$3.74 billion for agricultural production losses, including (but not limited to) losses due to drought. Of this amount, \$494.5 million is specifically to compensate livestock owners for losses in CY2022 due to qualifying drought and wildfires.

## **Disaster Relief Supplemental Appropriations Act, 2025**

Disaster Relief Supplemental Appropriations Act, 2025 (passed as Division B of the American Relief Act, 2025 [P.L. 118-158]) included payments for agricultural losses in calendar year 2023 and 2024 from qualifying disaster events, including drought and extreme heat. Specifically, the 2025 supplemental appropriations provided \$30.78 billion for agricultural losses, including (but not limited to) losses from drought. Of this amount, \$2 billion is specifically to compensate livestock owners for losses due to qualifying drought, wildfires, and floods in 2023, 2024 or both calendar years.

## **Proposed Drought Authorities in the 119<sup>th</sup> Congress**

Drought events have increased the profile of drought and have led to congressional and administrative proposals to prepare for and respond to drought's effects. Congressional interest in drought has included new and amended authorities for drought planning, preparedness, and response, and oversight of ongoing federal efforts to address the impacts of drought.<sup>206</sup> The following sections discuss some drought-related legislative proposals considered by the 119<sup>th</sup> Congress.

### **Coordination**

Legislation introduced in the 119<sup>th</sup> Congress, such as H.R. 5610 and S. 904, would direct the Secretary of Agriculture to establish an interagency working group to “improve the availability of consistent, accurate, and reliable data for use in producing” the USDM, with a focus on including additional data. The proposed interagency working group would have representatives from USDA, NOAA, NDMC, DOI, and nonfederal partners if enacted.

### **Monitoring and Research**

Other legislation introduced in the 119<sup>th</sup> Congress focuses on water-related monitoring and research. Legislation introduced in the 119<sup>th</sup> Congress includes the following:

---

<sup>205</sup> Funding was specifically provided for the Environmental Quality Incentives Program, Conservation Stewardship Program, Agricultural Conservation Easement Program, Regional Conservation Partnership Program, Watershed Protection and Flood Prevention Operations program, Grassroots Source Water Protection Program, Voluntary Public Access and Habitat Incentives Program, and Feral Swine Eradication and Control Pilot program.

<sup>206</sup> New and amended authorities are described in the sections above.

- H.R. 3816, as introduced; H.R. 5089, as ordered to be reported (amended); and S. 3923, as ordered to be reported with amendments favorably, would make changes to NIDIS, including adding functions related to next-generation technology, observations, and research, and authorizing appropriations through FY2030. It also would direct the Under Secretary of Commerce (known as the NOAA Administrator) to (1) maintain the National Mesonet Program, a nonfederal environmental observational network, to improve understanding and forecasting of droughts, among other phenomena and activities; (2) with other department and agency heads of USDA, USGS, NASA, and others, support the development, deployment, and maintenance of soil moisture monitoring networks; and (3) establish pilot projects to improve subseasonal to seasonal precipitation forecasts for water management and agriculture in the certain parts of the United States. Other legislation, such as S. 324, as introduced, would also address subseasonal and seasonal precipitation forecasts.
- S. 1626, as passed by the Senate, and S. 3923, as ordered to be reported with amendments favorably, would extend authorization of appropriations for the USGS's Federal Priority Streamgage Program and National Groundwater Monitoring Network through 2033 and direct the USGS to prioritize such monitoring in regions where there is a high level of drought; and authorize USGS's Next Generation Water Observing System and direct the USGS to prioritize drought activities, including drought forecasts.
- H.R. 5610, as introduced, and H.R. 7567, as passed by the House, would authorize appropriations to the Secretary of Agriculture for the USDM through FY2030.
- H.R. 453, as introduced, would direct the Secretaries of Agriculture, Commerce, and the Interior to administer a Wildfire Intelligence Center and, through it, work with the private sector to improve drought monitoring and forecasting, among other activities.

## **NOAA Drought Support Programs**

With respect to fishery resource disaster assistance, H.R. 4800, as introduced in the 119<sup>th</sup> Congress, would make certain freshwater crawfish fisheries eligible for disaster assistance. Among its contents, the bill would require a requester to include information about drought and/or other environmental factors to support a disaster finding for those fisheries.

## **Reclamation Programs**

Several legislative proposals in the 119<sup>th</sup> Congress would reauthorize and/or amend existing authorities related to Reclamation and drought in the West:

- H.R. 9022, as reported, includes under Section 203 provisions that would extend and/or authorize additional funding to several WIIN Act drought authorities, including authorities for new water project construction, water reuse and recycling, and desalination.
- S. 3738, as ordered to be reported with an amendment in the nature of a substitute favorably, would among other things, authorize \$800 million in funding for water conveyance projects, as well as additional funding for water recycling and reuse projects and environmental restoration projects.

- H.R. 338 and S. 3732, as introduced, would amend and extend Reclamation's IJA authority for small water storage projects.
- H.R. 6641 (the Central Valley Water Solution Act), as introduced, would authorize Reclamation to provide financial and technical assistance for 22 individual surface and ground water storage and conveyance projects at a total cost of approximately \$4.4 billion, as well as selected water recycling projects.
- H.R. 10420, as introduced, would authorize a total of \$19 billion for expediting construction of water and energy infrastructure in the Colorado River Basin and provide federal financial assistance for water augmentation projects.

## **USACE Programs**

The House-passed Water Resources Development Act of 2026 (H.R. 9497) would amend USACE's small water storage project authority established in Section 155 of P.L. 116-260 to support studies and construction for drought resiliency, including small water storage projects. Under the amended authority, drought resiliency measures could include construction of small water storage projects, sediment management, aquifer recharge, planting of native species, mitigation and monitoring of nonnative species that exacerbate drought (e.g., salt cedar), and other actions that increase drought resiliency and water conservation.

## **EPA Programs**

In the 119<sup>th</sup> Congress, some Members introduced legislation that would revise certain EPA programs to address drought conditions, sometimes for specific areas, as well as create new programs. The Water Resources Development Act of 2026 (S. 4949, as reported with an amendment in the nature of a substitute) would reauthorize appropriations for the DWSRF, as well as for the Drinking Water System Infrastructure Resilience and Sustainability program, among other changes.<sup>207</sup> It would also establish a new grant program to support digital infrastructure technology. Under this grant program, EPA would provide grants to, among other entities, water systems or treatment works experiencing a critical water supply need for a range of digital infrastructure technology projects, including those focused on “real-time data acquisition” to support aquifer recharge. This bill would further clarify that, subject to a state's discretion, DWSRF assistance can be used to address emerging contaminants in communities located in drought-prone regions that rely on groundwater as a primary or secondary source of drinking water, including those identified by EPA as sole source aquifers. Another introduced bill, H.R. 7845, would allow EPA or USACE to prioritize WIFIA loans for states or counties in drought areas, or for drought-related projects. This bill would also increase the total maximum amount of federal assistance to 90%, from 80%, for a drought project or project in a drought area that receives a WIFIA loan. H.R. 8461, as introduced, would reauthorize appropriations for the Indian Reservation Drinking Water Grant Program, described above, at \$60 million for each of FY2027 and FY2028. This bill would also add 10 tribal water projects in the Upper Colorado River Basin to the list of projects eligible for this grant program.

---

<sup>207</sup> For more information about the sole source aquifer program, see CRS Report RL31243, *Safe Drinking Water Act (SDWA): A Summary of the Act and Its Major Requirements*, by Elena H. Humphreys.

## USDA Programs

The 119th Congress' consideration of a farm bill is ongoing, with both House and Senate proposals including reauthorizations and amendments to existing USDA drought authorities. The farm bill is an omnibus act that authorizes, reauthorizes, or amends food, agriculture, and rural policies. On April 30, 2026, the House passed H.R. 7567, the Farm, Food, and National Security Act of 2026, which would amend conservation, research, monitoring, and forestry authorities related to drought and water management. On August 6, 2026, the Senate Committee on Agriculture, Nutrition, and Forestry considered a draft bill—Agricultural Act of 2026 (no bill number assigned)—with amendments.<sup>208</sup> The Senate committee's draft bill contains changes to authorities related to drought and water management similar to H.R. 7567. The Senate committee's draft bill also contains amendments to existing USDA authorities, specifically those for water management and water conservation on private agricultural land.

## Author Information

Charles V. Stern, Coordinator  
Specialist in Natural Resources Policy

Anna E. Normand  
Analyst in Natural Resources Policy

Eva Lipiec, Coordinator  
Specialist in Natural Resource Policy

Pervaze A. Sheikh  
Specialist in Natural Resources Policy

Lisa S. Benson  
Analyst in Agricultural Policy

Megan Stubbs  
Specialist in Agricultural Conservation and Natural Resources Policy

Nicole T. Carter  
Specialist in Natural Resources Policy

Zachary T. Neuhofer  
Analyst in Agricultural Policy

Elena H. Humphreys  
Analyst in Environmental Policy

Anthony R. Marshak  
Analyst in Natural Resources Policy

Mariel J. Murray  
Specialist in Natural Resources Policy

---

<sup>208</sup> For text of the Senate draft bill and its approved amendments, see Senate Committee on Agriculture, Nutrition, and Forestry, "Legislation," <https://www.agriculture.senate.gov/hearings/business-meeting-08-06-2026>.

---

## **Disclaimer**

This document was prepared by the Congressional Research Service (CRS). CRS serves as nonpartisan shared staff to congressional committees and Members of Congress. It operates solely at the behest of and under the direction of Congress. Information in a CRS Report should not be relied upon for purposes other than public understanding of information that has been provided by CRS to Members of Congress in connection with CRS's institutional role. CRS Reports, as a work of the United States Government, are not subject to copyright protection in the United States. Any CRS Report may be reproduced and distributed in its entirety without permission from CRS. However, as a CRS Report may include copyrighted images or material from a third party, you may need to obtain the permission of the copyright holder if you wish to copy or otherwise use copyrighted material.